

Waves and Oscillation

Periodic Motion: Motion that repeats itself its path after regular intervals of time is called periodic motion.

ex:-

- i) Motion of planets around the sun
- ii) Motion of electrons around the nucleus.

Oscillatory Motion: The motion of a body is said to be oscillatory if it moves to and fro or back and forth about the mean position or equilibrium position after regular intervals of time. It also known as (vibratory motion).

ex:-

- i) To and fro motion of bob in pendulum.
- ii) Mass attached to spring.

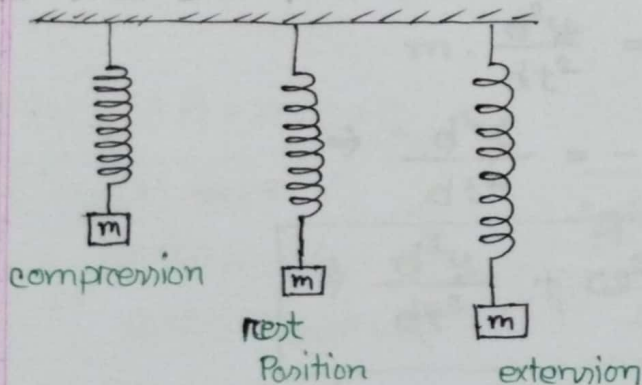


fig- mass-spring

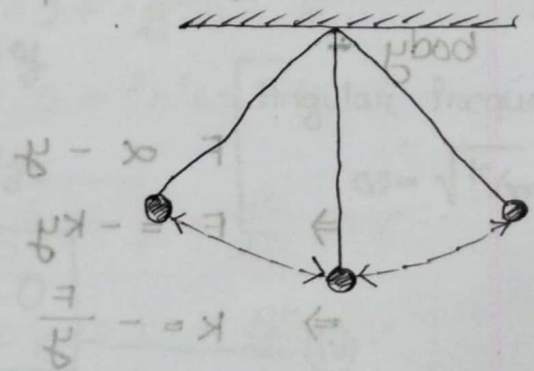


fig - pendulum

Simple Harmonic Motion (S.H.M): A body is said to be executing simple harmonic motion, if it moves to & fro about the mean position such that its acceleration is directly proportional to the displacement of the body from the mean position and direction of acceleration is opposite to the direction of displacement of the body.

ex:-

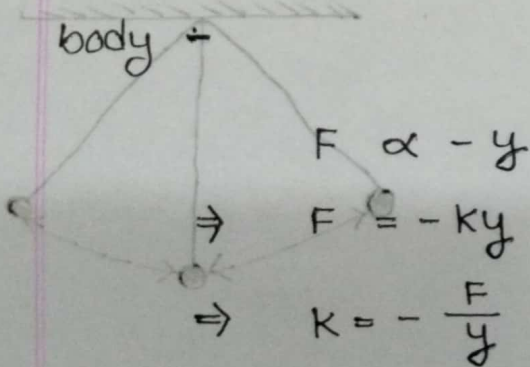
i) mass-spring.

Restoring force: The force which is developed in the body when it is displaced from the mean position is called restoring force.

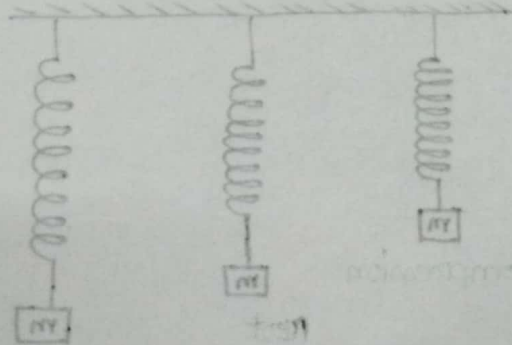
$$\text{restoring force} = - (\text{Applied force})$$

$$F = - F_{\text{ext}}$$

External force is proportional to the displacement of the body



k is spring constant.



Differential equation of SHM and its solution:

consider a body of mass 'm' moving to & fro motion about mean position 'O'. Let 'F' is restoring force and 'y' is displacement.

[restoring force]

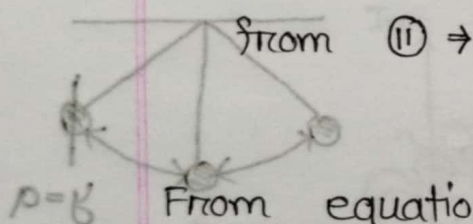
From Hooke's Law,

$$F = -ky \quad \text{--- (i)}$$

From Newton's second Law,

$$F = ma \quad \text{--- (ii)}$$

but acceleration $a = \frac{d^2y}{dt^2}$



From equation (i) & (ii)

$$F = m \cdot \frac{d^2y}{dt^2}$$

$$m \cdot \frac{d^2y}{dt^2} = -ky$$

$$\Rightarrow \frac{d^2y}{dt^2} = -\frac{k}{m} \cdot y$$

$$\Rightarrow \frac{d^2y}{dt^2} + \omega^2 y = 0$$

Angular frequency

$$\omega = \sqrt{k/m}$$

This is the differential equation
2nd order Linear Homogeneous equation
after solving this we get y.

[restoring force] $(\phi + \pi) = \frac{y}{A}$

Solve:

$$\textcircled{1} \times 2 \frac{dy}{dt} \Rightarrow$$

$$2 \frac{dy}{dt} \cdot \frac{d^2y}{dt^2} = -\omega^2 y \cdot 2 \frac{dy}{dt}$$

$$\Rightarrow \left(\frac{dy}{dt} \right)^2 = -\omega^2 y^2 + c \quad [\text{after integration}]$$

$$\Rightarrow \boxed{v = \frac{dy}{dt} = \sqrt{-\omega^2 y^2 + c}} \rightarrow \text{constant of integration.}$$

by applying boundary condition we get the value of c ,

if $y = a$,

$$\frac{dy}{dt} = 0$$

$$\Rightarrow -\omega^2 a^2 + c = 0$$

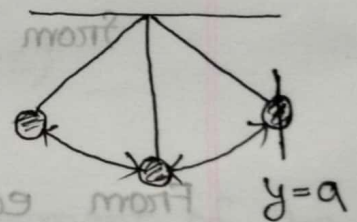
$$\Rightarrow \omega^2 a^2 = c$$

$$v = \frac{dy}{dt} = \sqrt{-\omega^2 y^2 + \omega^2 a^2}$$

$$\boxed{v = \frac{dy}{dt} = \pm \omega \sqrt{a^2 - y^2}}$$

$$\Rightarrow \frac{dy}{\sqrt{a^2 - y^2}} = \omega dt$$

$$\Rightarrow \sin^{-1} \frac{y}{a} = (\omega t + \phi) \quad [\text{after integration}]$$



$$* y = x = A \sin(\omega t + \phi)$$

$$v = \omega \sqrt{A^2 - x^2}$$

$$a = -\omega^2 x$$

$$v_{\max} = \omega A$$

$$a_{\max} = -\omega^2 A$$

$$\Rightarrow y = a \sin(\omega t + \phi) \rightarrow \text{solve of the differential eq.}$$

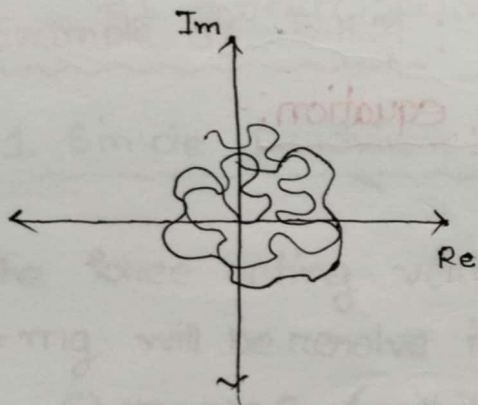
$$\Rightarrow y = a \sin \omega t \cos \phi + a \cos \omega t \sin \phi$$

$$= (a \cos \phi) \sin \omega t + (a \sin \phi) \cos \omega t$$

$$\Rightarrow y = A \sin \omega t + B \cos \omega t$$

$$\Rightarrow y = \text{Real} [A e^{i(\omega t + \phi)}]$$

$$\Rightarrow y(t) = \text{Real} [A \cos(\omega t + \phi) + i f(t)] \quad i = \sqrt{-1}$$

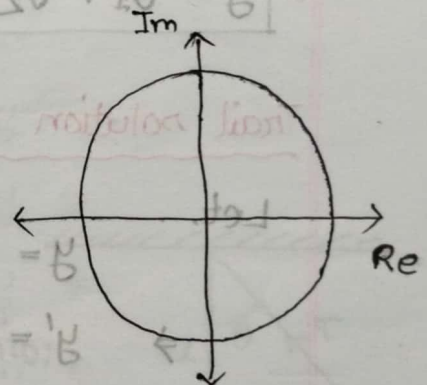


if $f(t) = \text{random / Arbitrary function}$

then graph could be anything

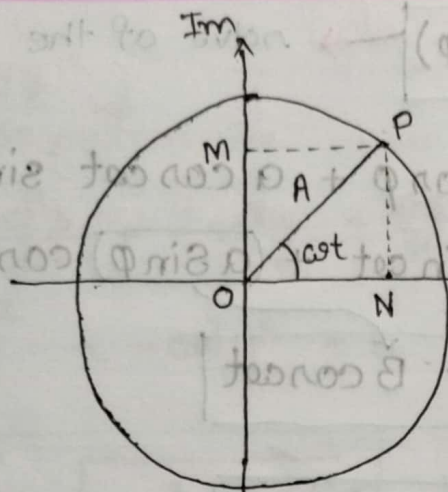
if $f(t) = \text{Periodic function}$

then graph will be a circle.



$$0 = \text{Im} (\omega + \text{Im}) \Leftarrow$$

$$0 = \omega + \text{Im} \Leftarrow$$



$$y_1 = ON = A \cos \omega t$$

$$y_2 = OM = A \sin \omega t$$

$$\therefore y = y_1 + y_2$$

Trail solution of differential equation:-

Let,

$$y = e^{mt}$$

$$\Rightarrow y' = m \cdot e^{mt}$$

$$\Rightarrow y'' = m^2 \cdot e^{mt}$$

Putting this values in differential equation of S.H.M.

$$m^2 e^{mt} + \omega^2 e^{mt} = 0$$

$$\Rightarrow (m^2 + \omega^2) e^{mt} = 0$$

$$\Rightarrow m^2 + \omega^2 = 0$$

$$\Rightarrow m^2 = -\omega^2$$

$$\Rightarrow m = \pm i\omega$$

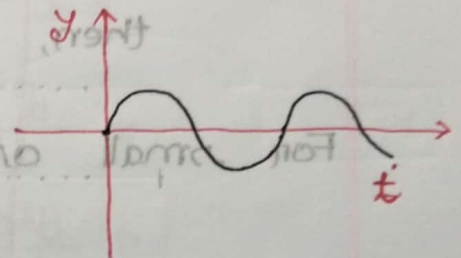
So,

$$y_1 = e^{i\omega t}$$

$$y_2 = e^{-i\omega t}$$

$$y = e^{i\omega t} + e^{-i\omega t}$$

$$\begin{cases} e^{i\theta} = \cos\theta + i\sin\theta \\ e^{-i\theta} = \cos\theta - i\sin\theta \end{cases}$$



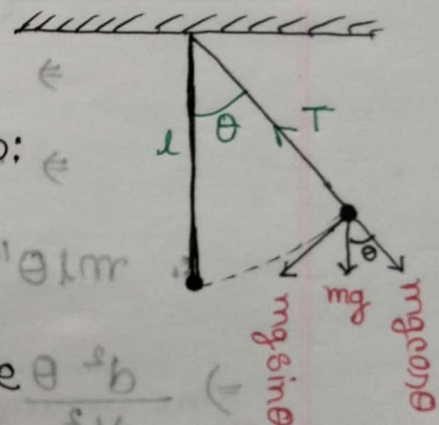
Example of S.H.M:

* 1. Simple pendulum:

The force acting vertically downward $= mg$ will be resolved into two components:

(i) $mg \cos\theta$ (will be balanced by Tension, T)

(ii) $mg \sin\theta$ (the only restoring force which will try to bring the pendulum to its mean position.)



$$0 = \frac{d^2\theta}{dt^2} + \frac{g}{l}\theta$$

$$\therefore F_R \propto -mg \sin \theta$$

From Maclaurin's series,

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots$$

if θ is very small,

$$\text{then, } \sin \theta = \theta$$

For small oscillation,

$$\text{the restoring force} = +mg\theta$$

$$\text{linear displacement} = l\theta$$

$$\therefore y = l\theta$$

$$\Rightarrow y' = l\theta'$$

$$\Rightarrow y'' = l\theta''$$

$$\Rightarrow my'' = ml\theta''$$

$$\therefore ml\theta'' = -mg\theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} + \frac{g}{l} \theta = 0$$

$$\text{here } \omega = \sqrt{\frac{g}{l}}$$

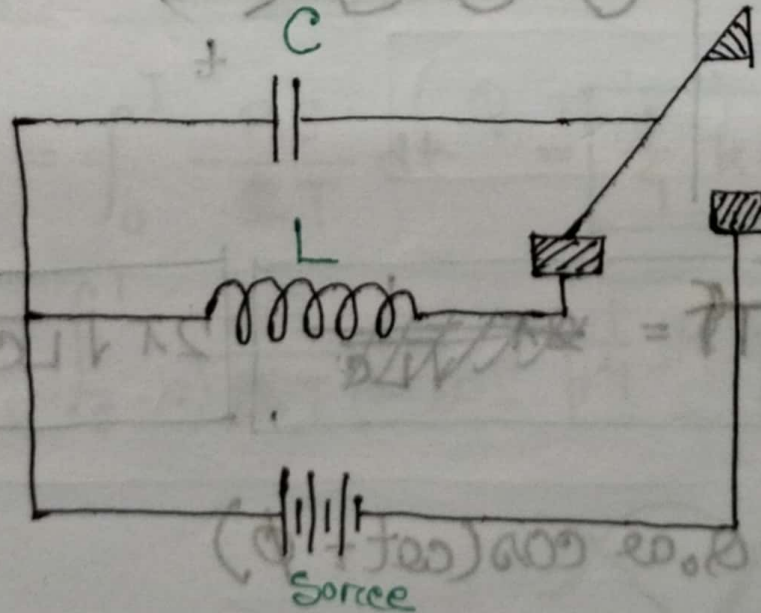
$$\Rightarrow \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0$$

It is very similar to S.H.M. equation.

Time period

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{L}{g}}$$

* 2. L-C circuit :-



The voltage across the capacitor $V_c = \frac{Q}{C}$

E.M.F across the inductor, $= -L \frac{di}{dt}$

Applying KVL,

$$L \frac{di}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d^2 Q}{dt^2} + \frac{Q}{C} = 0$$

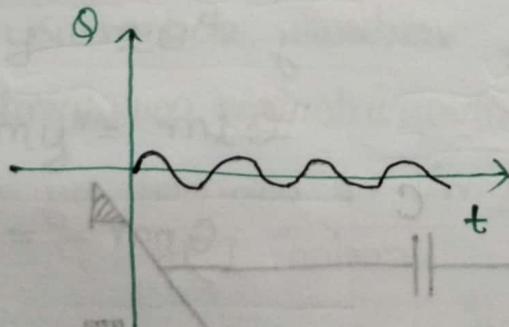
$$\Rightarrow \frac{d^2 Q}{dt^2} + \frac{Q}{LC} = 0$$

$$\Rightarrow \frac{d^2 Q}{dt^2} + \omega^2 Q = 0$$

$$\omega = \frac{1}{\sqrt{LC}}$$

By solving the equation,

$$Q = Q_0 \sin(\omega t + \phi)$$

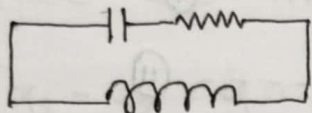


Time period
Time period,

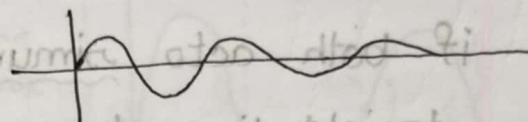
$$T = \frac{1}{\omega} = 2\pi \sqrt{LC}$$

$$i = \frac{dQ}{dt} = Q_0 \omega \cos(\omega t + \phi)$$

if circuit has resistance,



$$\frac{d^2\theta}{dt^2} + \omega^2\theta + iR = 0$$



amplitude কমছে।

Energy of a simple harmonic oscillator :

In the absence of any non-conservative forces the mechanical energy of SHO =

$$= K.E + P.E$$

$$= \frac{1}{2}mv^2 + \frac{1}{2}ky^2$$

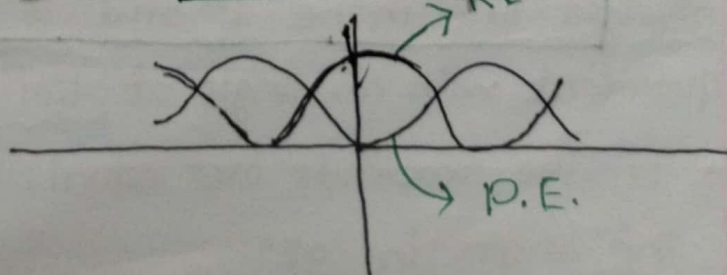
$$= \frac{1}{2}ma^2\omega^2 [\sin^2(\omega t + \phi) + \cos^2(\omega t + \phi)]$$

$$= \frac{1}{2}ka^2 = \text{constant}$$

$$\left[\begin{aligned} \omega &= \sqrt{\frac{k}{m}} \\ \Rightarrow k &= m\omega^2 \end{aligned} \right]$$

$$* \text{Avg. P.E} = \int_0^T \frac{P.E}{T} dt = \frac{1}{4}ka^2$$

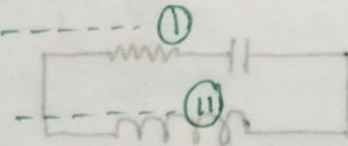
$$\text{Avg. K.E} = \int_0^T \frac{K.E}{T} dt = \frac{1}{4}ka^2$$



composition of SHM:

$$y_1 = a_1 \sin(\omega t + \alpha_1)$$

$$y_2 = a_2 \sin(\omega t + \alpha_2)$$



if both act simultaneously as a particle in a straight line, then the resulting displacement of the particle is the vector sum of individual displacement.

$$y = y_1 + y_2$$

$$= a_1 \sin(\omega t + \alpha_1) + a_2 \sin(\omega t + \alpha_2)$$

$$= a_1 [\sin \omega t \cos \alpha_1 + \cos \omega t \sin \alpha_1] +$$

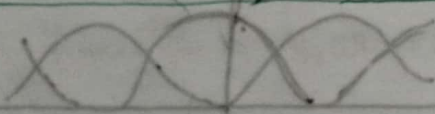
$$a_2 [\sin \omega t \cos \alpha_2 + \cos \omega t \sin \alpha_2]$$

$$= (a_1 \cos \alpha_1 + a_2 \cos \alpha_2) \sin \omega t + (a_1 \sin \alpha_1 + a_2 \sin \alpha_2) \cos \omega t$$

$$= A \cos \phi \sin \omega t + A \sin \phi \cos \omega t$$

$$= A \sin(\omega t + \phi)$$

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos(\alpha_1 - \alpha_2)}$$



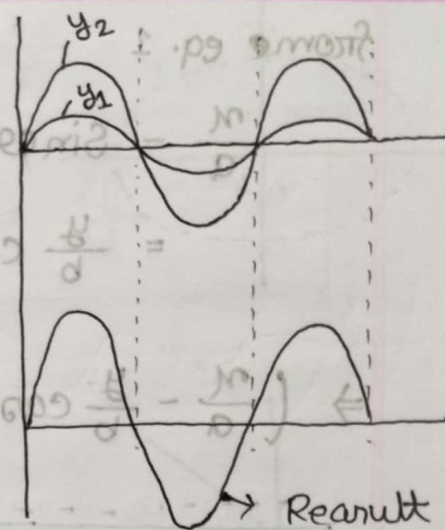
Now,

If particle are in same phase.

$$\alpha_1 = \alpha_2 = \alpha$$

$$\therefore A = a_1 + a_2$$

Same phase

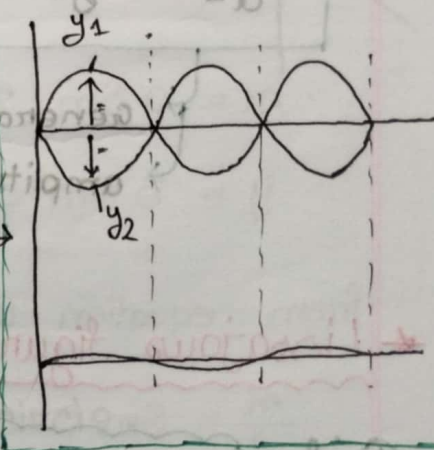


If particle are in opposite phase.

$$\alpha_1 - \alpha_2 = (2n+1)\pi$$

$$\therefore A = a_1 - a_2$$

Opposite phase



Now, if two perpendicular simple harmonic motion simultaneously act on a particle of a medium with same frequency but different amplitude and phase.

$$x = a \sin(\omega t + \phi) \quad \text{--- (i)}$$

$$y = b \sin \omega t \quad \text{--- (ii)}$$

P.T.O.

from eq. 1

$$\frac{x}{a} = \sin \omega t \cos \phi - \cos \omega t \sin \phi$$

$$= \frac{y}{b} \cos \phi - \sqrt{1 - \frac{y^2}{b^2}} \sin \phi$$

$$\Rightarrow \left(\frac{x}{a} - \frac{y}{b} \cos \phi \right)^2 = \left(1 - \frac{y^2}{b^2} \right) \sin^2 \phi$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} \cos \phi = \sin^2 \phi$$

General equation of conic.

amplitude depends on phase difference.

Lissajous figure: [Time period ratio 1:1]

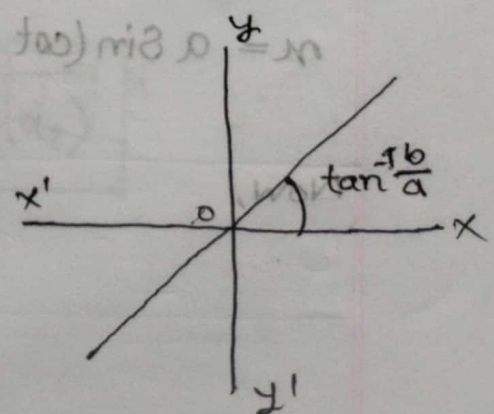
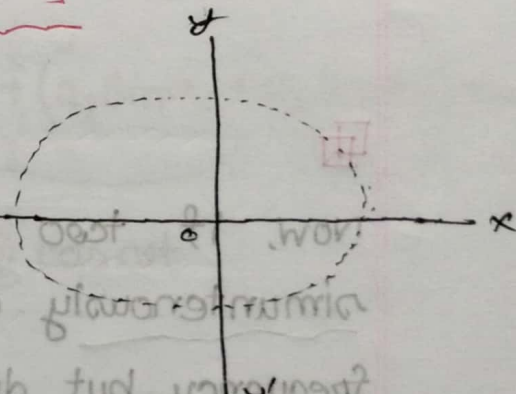
① if, $\phi = 0, 2\pi, 4\pi, \dots, 2n\pi$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} = 0$$

$$\Rightarrow y = \pm \frac{b}{a} x$$

This incident represent a pair of straight line in the 1st and 3rd quadrant with an angle of $\tan^{-1} \frac{b}{a}$.

* if the amplitude are equal, then the angle is 45°

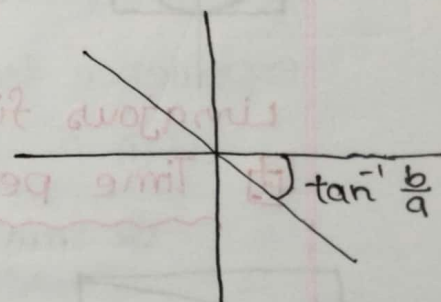
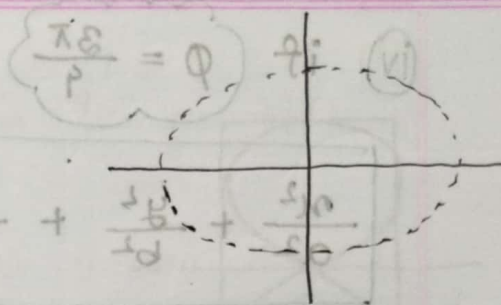


② if $\phi = \pi$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{2xy}{ab} = 0$$

$$y = \pm \frac{b}{a} x$$

This incident represent a pair of straight line in the 2nd and 4th quadrant with an angle of $\tan^{-1} \frac{b}{a}$.



③ if $\phi = \frac{\pi}{2}$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

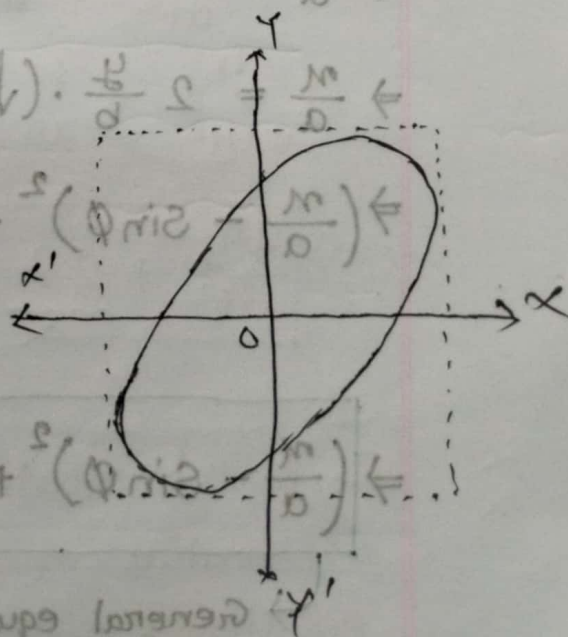
∴ if $a = b$, the equation will be a circle.

④ if $\phi = \frac{\pi}{4}$

$$\sin \phi = \cos \phi = \frac{1}{\sqrt{2}}$$

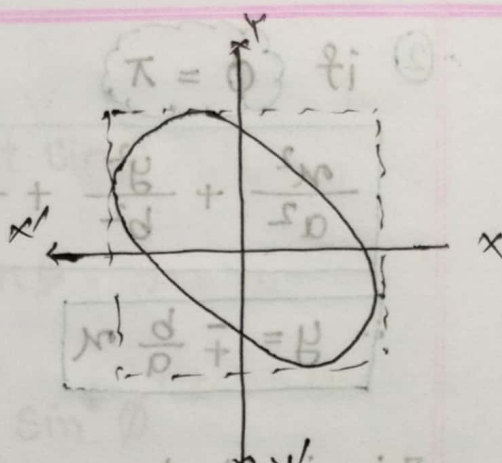
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{\sqrt{2} xy}{ab} = \frac{1}{2}$$

Equation of a oblique ellipse



(iv) if $\phi = \frac{3\pi}{4}$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{\sqrt{2}xy}{ab} = \frac{1}{2}$$



Limaçon figure

Time period ratio 1:2

$$x = a \sin(2\omega t + \phi) \quad \text{--- (i)}$$

$$y = b \sin \omega t \quad \text{--- (ii)}$$

from equation - (i)

$$\frac{x}{a} = \sin 2\omega t \cos \phi + \cos 2\omega t \sin \phi$$

$$\Rightarrow \frac{x}{a} = 2 \sin \omega t \cos \omega t \cos \phi + (1 - 2 \sin^2 \omega t) \sin \phi$$

$$\Rightarrow \frac{x}{a} = 2 \frac{y}{b} \cdot \left(\sqrt{1 - \frac{y^2}{b^2}} \right) \cdot \cos \phi + \left(1 - \frac{2y^2}{b^2} \right) \sin \phi$$

$$\Rightarrow \left(\frac{x}{a} - \sin \phi \right)^2 + \frac{4y^4}{b^4} \sin^2 \phi + 2 \left(\frac{x}{a} - \sin \phi \right)^2 \frac{2y^2}{b^2} \sin^2 \phi = \frac{4y^2}{b^2} \cos^2 \phi \sqrt{1 - \frac{y^2}{b^2}}$$

$$\Rightarrow \left(\frac{x}{a} - \sin \phi \right)^2 + \frac{4y^2}{b^2} \left[\frac{y^2}{b^2} + \frac{x}{a} \sin \phi - 1 \right] = 0$$

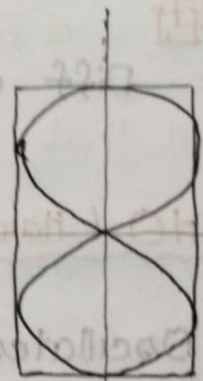
General equation of curve having two loops.

Note,

(i) if $\phi = 0$,

$$\frac{x^2}{a^2} + \frac{4y^2}{b^2} \left(\frac{y^2}{b^2} - 1 \right) = 0$$

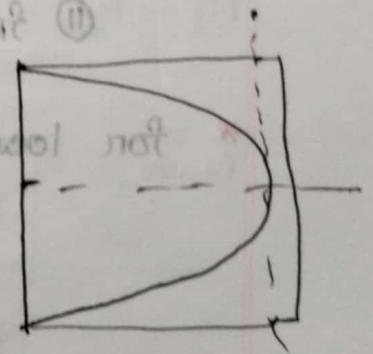
Resultant figure will represent the figure of eight.



(ii) if $\phi = \frac{\pi}{2}$,

$$y^2 = -\frac{b^2}{2a}(x-a)$$

Resultant figure will be



* Practice

$\frac{\pi}{4}, \pi$

$$0 = \frac{\mu}{b} \frac{p}{m} + \frac{\mu b}{b} \frac{p}{m} + \frac{\mu^2 b}{2b} m$$

$$0 = \frac{\mu}{b} \frac{p}{m} + \frac{\mu b}{b} \frac{p}{m} + \frac{\mu^2 b}{2b} m$$

$$0 = \mu^2 \omega + \frac{\mu b}{b} \omega + \frac{\mu^2 b}{2b} m$$

$$\omega = \sqrt{\frac{p}{m}}$$

This is the equation of 2nd order differential equation or damped equation.



Diff eq. of SHM : $\frac{d^2y}{dt^2} + \omega^2 y = 0$ (1)

HO (Harmonic Oscillation)

Oscillates in a damping or resistive medium and will experience two kinds of forces,

(i) Restoring force

(ii) frictional force

* for low speed:

frictional force $\propto$ velocity.

$$\Rightarrow F_f \propto \frac{dy}{dt}$$

then, the equation becomes:

$$m \frac{d^2y}{dt^2} + b \frac{dy}{dt} + \frac{a}{m} y = 0$$

$$\Rightarrow \frac{d^2y}{dt^2} + \frac{b}{m} \frac{dy}{dt} + \frac{a}{m} y = 0$$

$$\Rightarrow \frac{d^2y}{dt^2} + 2\lambda \frac{dy}{dt} + \omega^2 y = 0$$

λ = damping factory/co-efficient

$$\omega = \sqrt{\frac{a}{m}}$$

Solve এর সুবিধার জন্য 2λ ধরা হয়েছে

This is the equation of 2nd order differential equation or damped equation.

Trial Solution:

Let, $y = A e^{kt}$
 $y' = A k e^{kt}$
 $y'' = A k^2 e^{kt}$

from eq. $\rightarrow$

$$A k^2 e^{kt} + 2\lambda A k e^{kt} + \omega^2 A e^{kt} = 0$$

$$\Rightarrow (k^2 + 2\lambda k + \omega^2) \cdot A e^{kt} = 0$$

if $A e^{kt} \neq 0$, then

$$k^2 + 2\lambda k + \omega^2 = 0$$

$$\Rightarrow k = \frac{-2\lambda \pm \sqrt{4\lambda^2 - 4\omega^2}}{2}$$

$$\Rightarrow k = -\lambda \pm \sqrt{\lambda^2 - \omega^2}$$

So, $\therefore y_1 = A_1 e^{(-\lambda + \sqrt{\lambda^2 - \omega^2})t} \rightarrow \text{sol}^n 1$

$\therefore y_2 = A_2 e^{(-\lambda - \sqrt{\lambda^2 - \omega^2})t} \rightarrow \text{sol}^n 2$

Now, the general equation:

$$y = y_1 + y_2$$

$$y = A_1 e^{(-\lambda + \sqrt{\lambda^2 - \omega^2})t} + A_2 e^{(-\lambda - \sqrt{\lambda^2 - \omega^2})t}$$

P.T.O.

$$a_0 = A_1 + A_2 \quad \text{--- (1)}$$

at that condition velocity will be 0.

$$\therefore \frac{dy}{dt} = 0$$

So,

$$0 = (-\lambda + \sqrt{\lambda^2 - \omega^2}) A_1 e^{(-\lambda + \sqrt{\lambda^2 - \omega^2})t} + (-\lambda - \sqrt{\lambda^2 - \omega^2}) A_2 e^{(-\lambda - \sqrt{\lambda^2 - \omega^2})t}$$

again,

$$A_1 - A_2 = \frac{\lambda a_0}{\sqrt{\lambda^2 - \omega^2}}$$

Now,

$$A_1 = \frac{1}{2} a_0 \left(1 + \frac{\lambda}{\sqrt{\lambda^2 - \omega^2}} \right)$$

$$A_2 = \frac{1}{2} a_0 \left(1 - \frac{\lambda}{\sqrt{\lambda^2 - \omega^2}} \right)$$

Now, we will compare λ with ω .

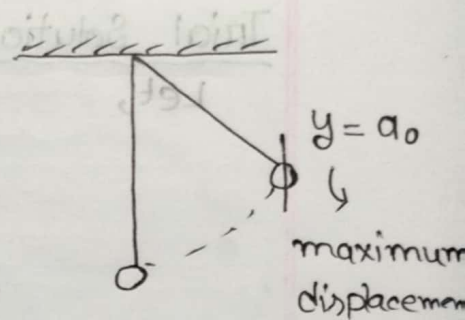
So,

The equation of damped oscillation:

$$\frac{d^2y}{dt^2} + 2\lambda \frac{dy}{dt} + \omega^2 y = 0$$

Soln.

$$y = A_1 e^{(-\lambda + \sqrt{\lambda^2 - \omega^2})t} + A_2 e^{(-\lambda - \sqrt{\lambda^2 - \omega^2})t}$$



at maximum displacement, velocity zero & $t=0$

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The effect of damping on displacement:

(i) $\lambda > \omega$

(ii) $\lambda < \omega$

(iii) $\lambda = \omega$

case 1: ($\lambda > \omega$) (Heavy damping):

* both side of the right hand side of the solⁿ eq.

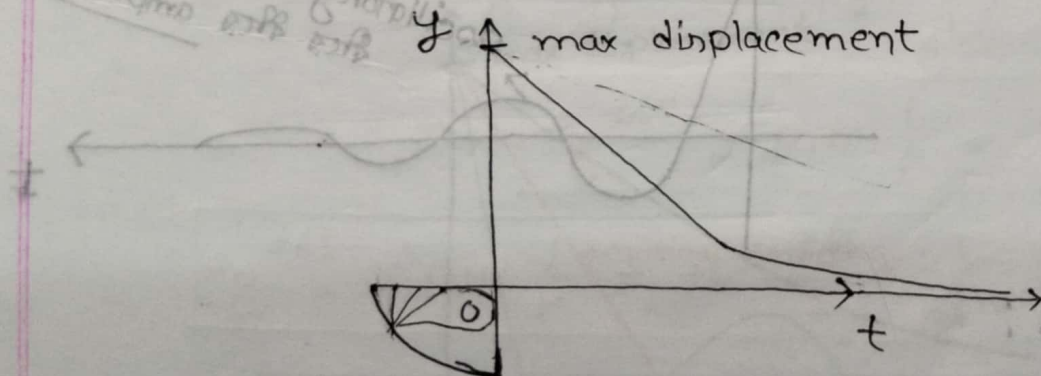
will be real quantity.

* so the nature of the solution will be non-oscillatory

* The displacement will decrease monotonically with

the increase of time and the system distracted

from the equilibrium position will return to equilibrium position ~~asymptotically~~ asymptotically, as $t \rightarrow \infty$



ex - door damper,

Case-2: ($\lambda < \omega$) (low damping / small damping):

both the term of right hand side of the equation will be imaginary quantity.

So the nature of the displacement will be oscillatory.

$$\sqrt{\lambda^2 - \omega^2} = i g = i \sqrt{\omega^2 - \lambda^2} \quad \left[i = \sqrt{-1} \right]$$

from soln eq:

$$y = e^{-\lambda t} (A_1 e^{igt} + A_2 e^{-igt})$$

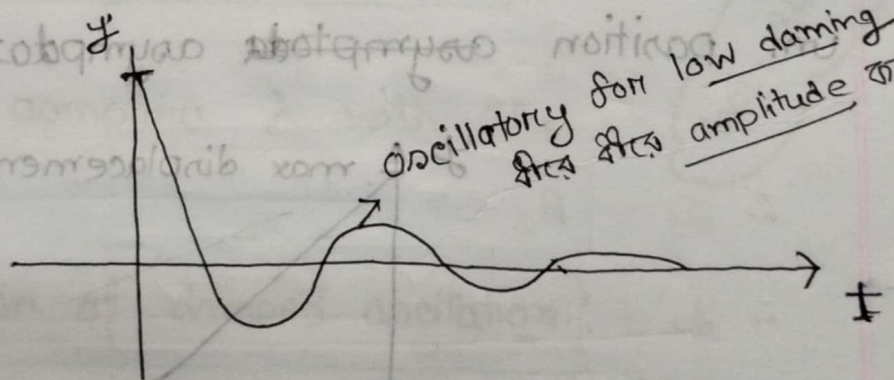
$g = \text{frequency of damped oscillator}$

$$\Rightarrow y = e^{-\lambda t} [(A_1 + A_2) \cos gt + i (A_1 - A_2) \sin gt]$$

$$\Rightarrow y = Q_0 e^{-\lambda t} \sin (gt + \phi)$$

$Q_0 e^{-\lambda t} = \text{Amplitude}$

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Example:

Vehicle suspension system.

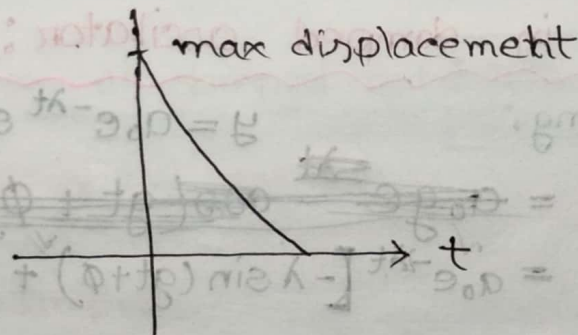
Case-3 ($\lambda = \omega$) (critical damping)

$$y = (A_1 + A_2)e^{-\lambda t}$$

So the displacement will be non-oscillatory.

depending on damping factor (λ) the system comes to rest in a very short time.

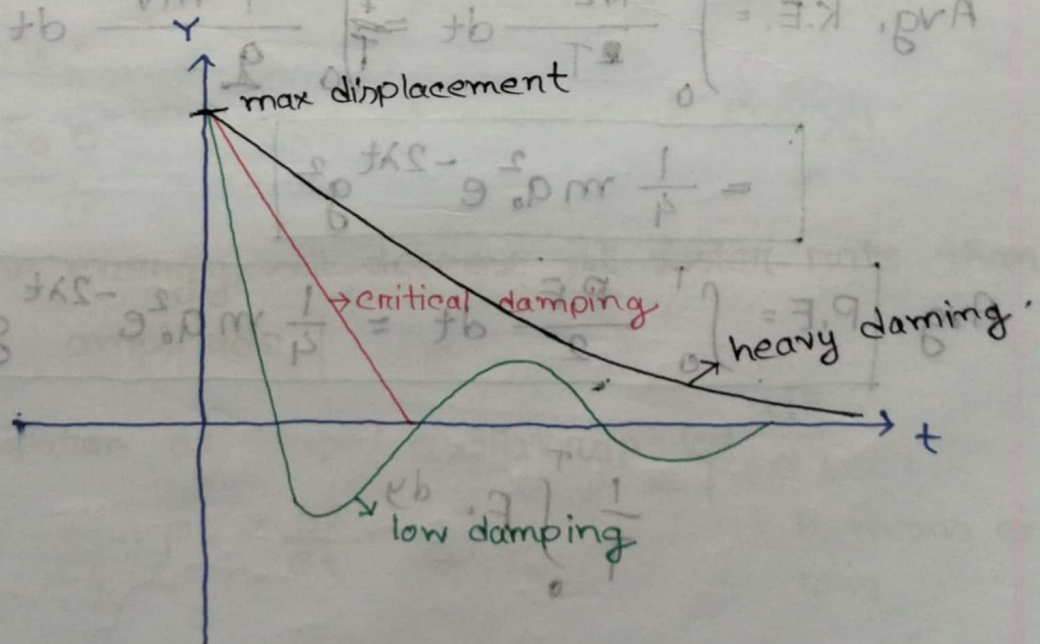
$\sqrt{\lambda^2 - \omega^2} = h$, where h is very small.



example:

- ① Pointing instrument
- ② Galvanometer
- ③ electronic feedback system
- ④ highly professional camera

**** Now all damping in one graph:-**



when $\lambda < \omega$,

$$y = a_0 e^{-\lambda t} \sin(gt + \phi)$$

Here,

$g \rightarrow$ frequency of damped oscillator
 $\rightarrow \sqrt{\omega^2 - \lambda^2}$
 $\rightarrow g$ is always less than natural frequency.

Energy

Energy loss in damped oscillator:-

for low damping,

$$y = a_0 e^{-\lambda t} \sin(gt + \phi)$$

$$v = \frac{dy}{dt} = \cancel{a_0 g e^{-\lambda t} \cos(gt + \phi)} + \cancel{a_0 e^{-\lambda t} (-g) \sin(gt + \phi)}$$

$$= a_0 e^{-\lambda t} [-\lambda \sin(gt + \phi) + g \cos(gt + \phi)]$$

We know,

$$K.E. = \frac{1}{2} m v^2$$

$$\text{Avg. K.E.} = \int_0^T \frac{K.E.}{T} dt = \frac{1}{T} \int_0^T \frac{1}{2} m v^2 dt =$$

$$= \frac{1}{4} m a_0^2 e^{-2\lambda t} g^2$$

$$\text{Avg. P.E.} = \int_0^T \frac{P.E.}{2} dt = \frac{1}{4} m a_0^2 e^{-2\lambda t} g^2$$

$$\frac{1}{T} \int_0^T F \cdot dy$$

So, At

$$\text{Total average energy} = K.E_{\text{avg}} + P.E_{\text{avg}}$$

$$= \frac{1}{4} m a_0^2 \omega^2 e^{-2\lambda t} + \frac{1}{4} m a_0^2 \omega^2 e^{-2\lambda t}$$

$$= \frac{1}{2} m a_0^2 \omega^2 e^{-2\lambda t}$$

$$= E_0 e^{-2\lambda t}$$

$$\therefore E_0 = \frac{1}{2} m a_0^2 \omega^2$$

when, $t = 0$
Energy of the undamped oscillator.

Power dissipation of damped oscillation:

$$y = a \sin(\omega t + \phi)$$

when, it is damped,

$$y = a_0 e^{-\lambda t} \sin(\omega t + \phi)$$

So, $a = a_0 e^{-\lambda t}$ → rate of decrease of amplitude

$E = E_0 e^{-2\lambda t}$ → rate of decrease of energy.

[N.B: The energy will decrease at faster rate than amplitude]

Power dissipation of damped oscillator $p = - \frac{dE}{dt}$

$p = - \frac{dE}{dt}$

it means energy loss.

Quality factor (Q):

$$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per period}}$$

$$\Rightarrow Q = 2\pi \frac{E}{\frac{2\lambda E T}{T}} \quad \left[T = \frac{2\pi}{\omega} \right]$$

$$\Rightarrow Q = \frac{\omega}{2\lambda}$$

$$\Rightarrow Q = \frac{\omega}{2\lambda}$$

[N.B:] The less the damping the better will be the quality factor.

usually,

$Q < \frac{1}{2}$ for overdamped system

$Q > \frac{1}{2}$ for underdamped system

$Q = \frac{1}{2}$ for critical damping system.

Amplitude of a underdamped oscillator,

$$a_t = a_0 e^{-\lambda t}$$

Now, if at $a_t \Rightarrow \frac{a_0}{e}$, then,

$$\frac{a_0}{e} = a_0 e^{-\lambda \tau} \quad \text{"\tau" \rightarrow relaxation time.}$$

$$\tau = \frac{1}{\lambda}$$

$$\tau = \frac{1}{\lambda}$$

relaxation time:- The time required for the amplitude of a underdamped oscillator to reduce to $\frac{1}{e}$ th of its initial value.

$$\frac{d^2y}{dt^2} + \omega^2 y = 0 \longrightarrow \text{SHM}$$

$$\Rightarrow \left[\frac{d^2y}{dt^2} + 2\lambda \frac{dy}{dt} + \omega^2 y = 0 \right] \longrightarrow \text{Damped oscillation}$$

বাইরে থেকে external force apply করলে There will be puzzle between system's frequency and external forces frequency.

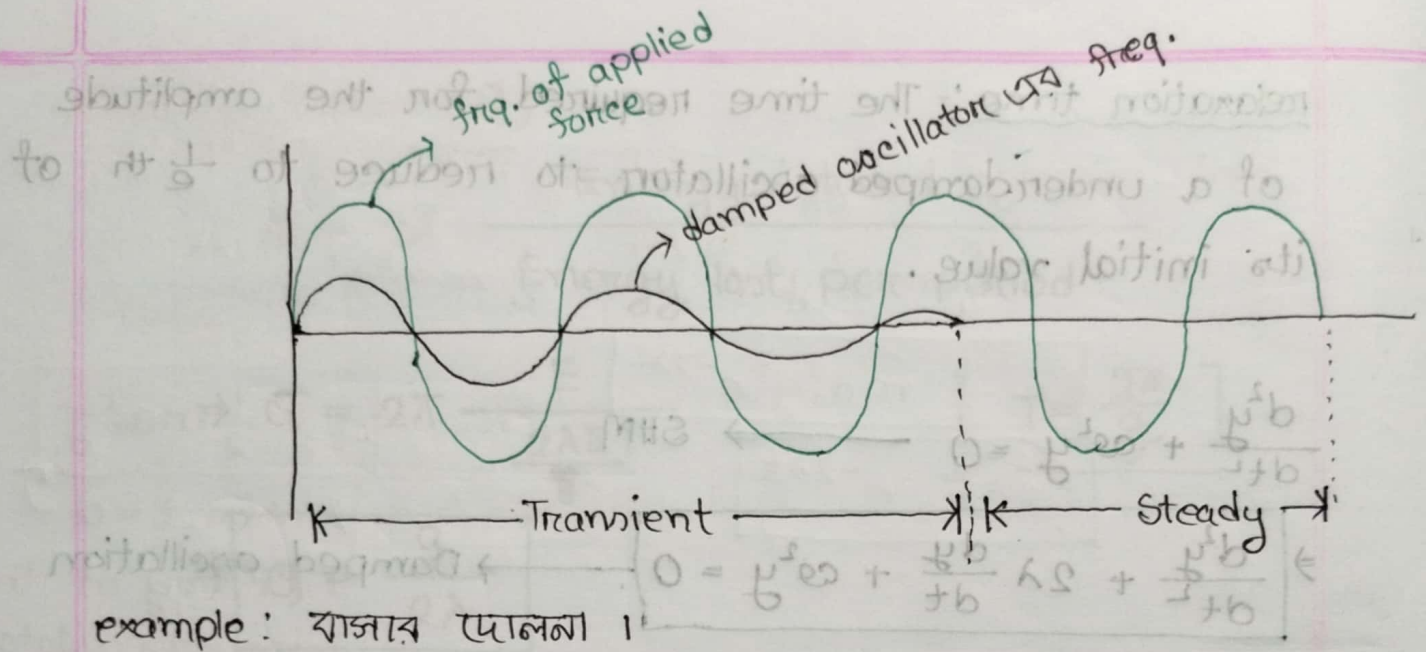
* Damped oscillation কে undamped করতে বাইরে থেকে external force periodic force apply করতে হয়।

as a result, কিছু সময় পর damping হারিয়ে মাত্র and periodic force ~~to~~ টাই থেকে মাত্র যা oscillation ঘটাবে।

① force vibration:-

Applied force $F = F_0 \sin pt$

In that case, $\frac{d^2y}{dt^2} + 2\lambda \frac{dy}{dt} + \omega^2 y = F$



proof:

Let the periodic function,

$$F = F_0 \sin pt$$

The eq. of motion of the force vibration

$$my'' + by' + ay = F_0 \sin pt$$

$$\Rightarrow y'' + \frac{b}{m} y' + \frac{a}{m} y = \frac{F_0}{m} \sin pt$$

$$\Rightarrow y'' + 2\lambda y' + \omega^2 y = f_0 \sin pt \quad \text{--- ①}$$

eq. of force vibration

2nd degree non-homogeneous diff. eq.

Solve:

Let particular solution of this eq.

$$y = A \sin(pt - \phi)$$

$$y'' = -Ap^2 \sin(pt - \phi)$$

$$y' = Ap \cos(pt - \phi)$$

Now,

$$\begin{aligned}
 -A p^2 \sin(pt - \phi) + 2\lambda A p \cos(pt - \phi) + \omega^2 A \sin(pt - \phi) &= \\
 &= f_0 \sin[(pt - \phi) + \phi] \\
 &= f_0 [\sin(pt - \phi) \cos \phi + \cos(pt - \phi) \sin \phi]
 \end{aligned}$$

Now, equating the co-efficient of this $\sin(pt - \phi)$ and $\cos(pt - \phi)$

$$A(\omega^2 - p^2) = f_0 \cos \phi \quad \text{--- (ii)} \quad [\sin(pt - \phi)]$$

$$2\lambda A p = f_0 \sin \phi \quad \text{--- (iii)} \quad [\cos(pt - \phi)]$$

from (ii) + (iii) $\Rightarrow$

$$A = \frac{f_0}{\sqrt{(\omega^2 - p^2)^2 + 4\lambda^2 p^2}}$$

* for a finite value of λ , A will be max when the denominator will have its min value.

* damping এর কারণে λ - less হলে তা heat আকারে release হবে।

Now, $y_c = a_0 e^{-\lambda t} \sin(gt + \phi)$

The general solution: $= y_p + y_c$

$$y = \underbrace{a_0 e^{-\lambda t} \sin(gt + \phi)}_{\text{Transient term}} + \underbrace{A \sin(pt - \phi)}_{\text{Steady state}}$$

3/1/2

Now,

$$= (\phi - \tan^{-1} \frac{d}{dp}) \left[(\omega^2 - p^2)^2 + 4\lambda^2 p^2 \right] = 0 + (\phi - \tan^{-1} \frac{d}{dp})$$

$$\Rightarrow -2(\omega^2 - p^2) \cdot 2p + 4\lambda^2 \cdot 2p = 0$$

$$\Rightarrow p = \sqrt{\omega^2 - 2\lambda^2}$$

$$\Rightarrow \boxed{\frac{p}{2\pi} = \frac{\sqrt{\omega^2 - 2\lambda^2}}{2\pi}}$$

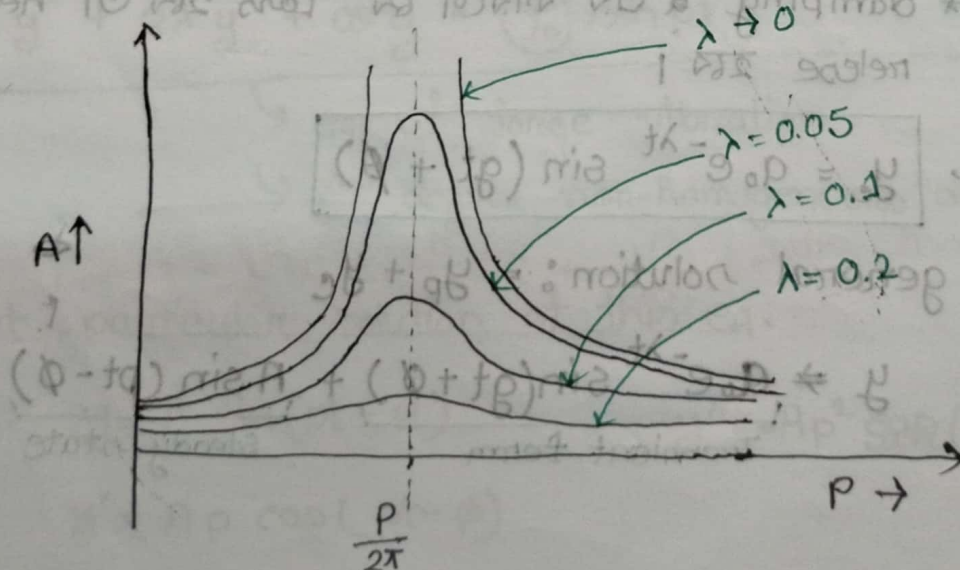
→ resonance frequency. → (ω) freq. ~~maximum~~
maximum.
→ occurs when ω = ω_{res} freq. max.

The freq

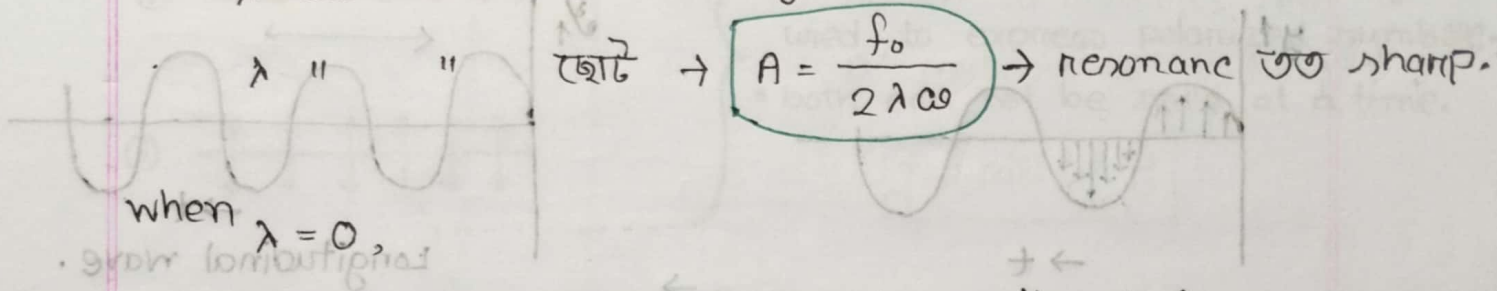
Now,

$$A = \frac{f_0}{\sqrt{4\lambda^2(\omega^2 - \lambda^2)}} = \frac{f_0}{2\lambda(\sqrt{\omega^2 - \lambda^2})}$$

$$\Rightarrow \boxed{A_{\max} = \frac{f_0}{\sqrt{4\lambda^2(\omega^2 - \lambda^2)} \cdot 2\lambda}$$



① λ এর value বেশি $\rightarrow$ graph flat



when

$$\lambda = 0$$

we can not design a device without λ . we need at least a value of λ , so, $\lambda = 0$ is not possible for practice.

example:

- ① music instrument
- ② washing machine
- ③ Human vocal cord

Wave motion

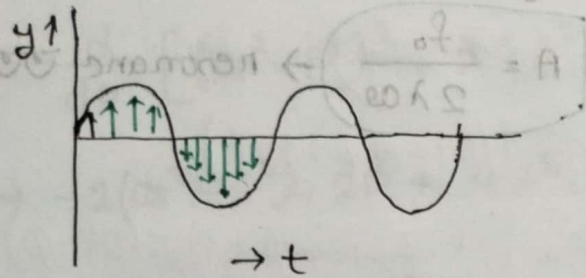
Mechanical motion

↓
Sound wave

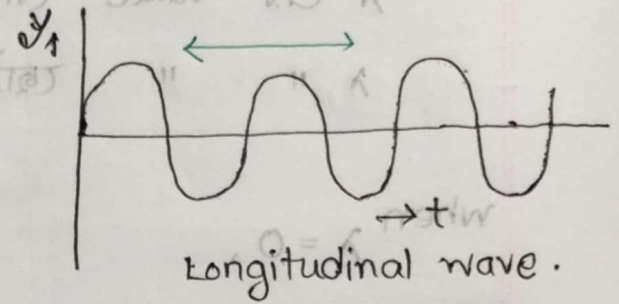
- * material medium is mandatory for sound wave transfer
- * transfer by disturbance
- * can be either transverse or Longitudinal

Electromagnetic wave / Non-mechanical wave
↓
light

- * medium is not necessary
- * Always transverse in nature.



Transverse wave

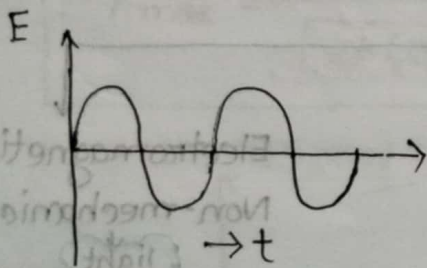


Longitudinal wave.

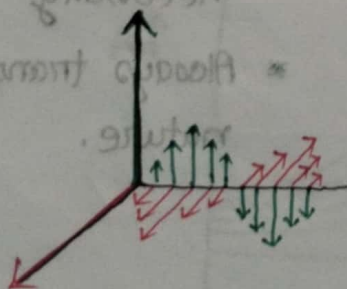
Direction of electromagnetic wave.

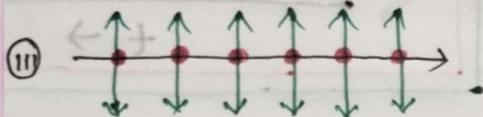
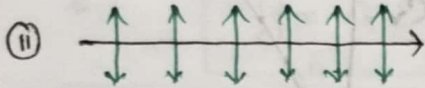
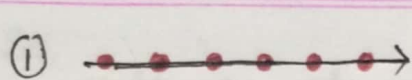
$\vec{E}$ & $\vec{H}$ both have same amplitude and must be perpendicular.

Mechanical wave motion :-



→ it's a plain polarized wave.
(in sinusoidal form)





① & ② are both individually used to express polarized symbols. both can not be used at a time.

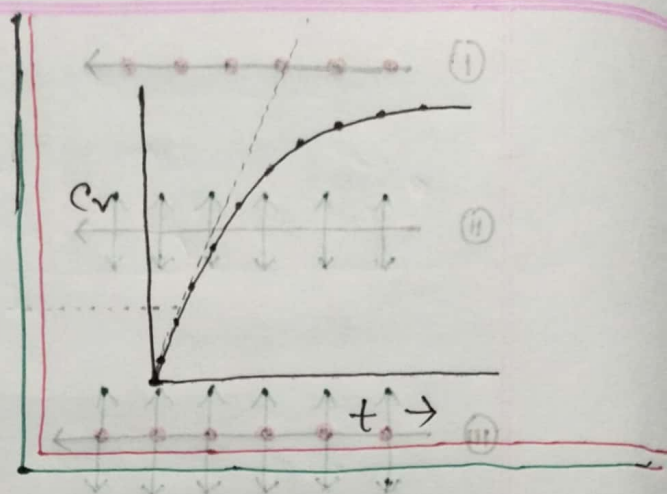
used to express unpolarized wave.

Sound wave motion:

- repeated periodic motion of the particle of the medium.
- only the wave travel forward while the particle of the medium will oscillate about their mean position.
- There will be a regular phase difference between two successive path difference of the medium.
- The velocity of wave is different from the velocity of particle of the medium which oscillates.
- wave velocity is constant for a particular medium.
- wave velocity lattice structure ~~dictates~~ dictates electromagnetic electronic property.

The equation of motion of the particle at position 'o'

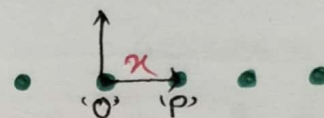
$$y = a \sin \omega t$$



The equation of motion of another particle at 'p' at a distance 'x' from 'o' in the positive -x direction

$$y = a \sin (\omega t - \phi)$$

Phase diff.



$$\Rightarrow y = a \sin \left(\omega t - \frac{2\pi}{\lambda} x \right)$$

$$\Rightarrow y = a \sin (\omega t - kx)$$

plane progressive wave

amplitude constant (without the attenuation of energy)

another form:

$$y = a \sin k(vt - x)$$

$$= a \sin \frac{2\pi}{\lambda} (vt - x)$$

constant phase $(\omega t - kx)$

• phase velocity: $\frac{d}{dt} (\omega t - kx) = 0 \Rightarrow \omega - k \frac{dx}{dt} = 0 \Rightarrow \frac{dx}{dt} = \frac{\omega}{k}$
 $\Rightarrow v\lambda = v \rightarrow$ velocity of wave

So, diff. eq. of wave motion:

$$y = a \sin \frac{2\pi}{\lambda} (vt - x) \rightarrow \text{eq. of progressive wave.}$$

particle velocity $y' = \frac{dy}{dt}$

$$= \frac{2\pi av}{\lambda} \cos \frac{2\pi}{\lambda} (vt - x)$$

acceleration $y'' = \frac{d^2y}{dt^2}$

$$= -\frac{4\pi^2 v^2 a}{\lambda^2} \sin \frac{2\pi}{\lambda} (vt - x)$$

with respect to 't'

again,

$$\text{velocity } y' = \frac{dy}{dx} = -\frac{2\pi}{\lambda} a \cos \frac{2\pi}{\lambda} (vt - x)$$

$$\text{acceleration } y'' = \frac{dy}{dx} = -\frac{2\pi^2}{\lambda^2} \cdot a \sin \frac{2\pi}{\lambda} (vt - x)$$

with respect to 'x'

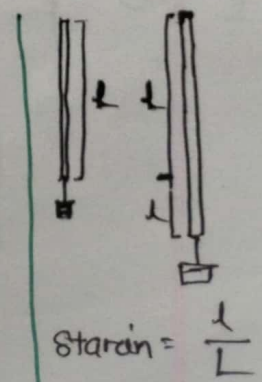
Now,

$$\frac{d^2y}{dt^2} = K \frac{d^2y}{dx^2}$$

→ diff. eq of wave motion (10)

$$\Rightarrow K = v^2$$

$$\Rightarrow v = \sqrt{K}$$



Energy :-

K.E. of plane progressive wave per unit volume:

$$K.E = \frac{1}{2} \rho v^2 = \frac{1}{2} \rho \frac{2\pi^2 a^2 v^2 \cos^2 \left[\frac{2\pi}{\lambda} (vt - x) \right]}{\lambda^2}$$

P.E of plane progressive wave per unit vol.

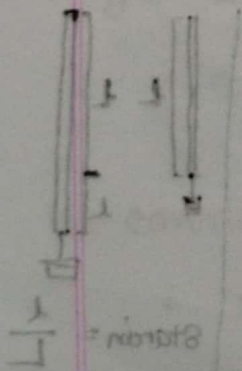
$$P.E = \omega = \int F \cdot dy = \int \rho \cdot \frac{du}{dt} \cdot dy$$

$$= \frac{2\pi^2 a^2 v^2}{\lambda} \rho \sin^2 \left[\frac{2\pi}{\lambda} (vt - x) \right]$$

So, Total energy = Kinetic energy + potential energy.

$$= \rho \cdot \frac{2\pi^2 a^2 v^2}{\lambda^2} \rightarrow n^2 [v = n\lambda]$$

So, Total energy is constant.



$$\Rightarrow v = \frac{\omega}{k}$$

$$\Rightarrow v = \frac{v}{k}$$

Stationary wave :-

* fixed boundary থাকবে।

* যেমন: e- ঘুরে নিউক্লিয়াসের চারপাশে।

* When two simple harmonic progressive wave of same frequency and same amplitude travel to a straight line but in opposite direction, form stationary wave.

Let consider,

direct wave : $y_1 = a \sin \frac{2\pi}{\lambda} (vt - x)$

reflected wave : $y_2 = -a \sin \frac{2\pi}{\lambda} (vt - x)$

Now, the resultant amplitude:

$$y = y_1 + y_2 \\ = a \sin \frac{2\pi}{\lambda} (vt - x) - a \sin \frac{2\pi}{\lambda} (vt - x)$$

$$= \underbrace{-2a \sin \frac{2\pi}{\lambda} \cdot x}_{A} \cos \frac{2\pi}{\lambda} vt$$

$$\left. \begin{array}{l} \sin C - \sin D \\ \Rightarrow 2 \cos \frac{C+D}{2} \\ \sin \frac{C-D}{2} \end{array} \right\}$$

Now,

Particle velocity,

$$u = \frac{dy}{dt} = \frac{4\pi av}{\lambda} \sin \frac{2\pi}{\lambda} x \cdot \sin \frac{2\pi vt}{\lambda}$$

Particle acceleration,

$$a = \frac{dy}{dt} = \frac{8\pi^2 v^2}{\lambda^2} a \sin \frac{2\pi}{\lambda} x \sin \frac{2\pi vt}{\lambda}$$

Now, when wave is simple progressive wave

Strain:

$$\frac{dy}{dx} = -\frac{4\pi a}{\lambda} \cos \frac{2\pi}{\lambda} x \cos \frac{2\pi vt}{\lambda}$$

Now,

if (কোনো) wave is simple progressive হলে তার amplitude constant হয়,

So, $y = A \sin \frac{2\pi}{\lambda} (vt - x)$

Now changing with respect to 'x'

Case 1: $\sin \frac{2\pi}{\lambda} = 0$, হলে, $\cos \frac{2\pi}{\lambda} x = \pm 1$ হবে,

and, $y = 0$, $v = 0$, $a = 0$, $A = 0$ হবে, কিন্তু

$$\frac{dy}{dx} = \text{maximum} \approx 1$$

It is only possible when,

$$x = \frac{m\lambda}{2} \quad [m = 1, 2, 3, \dots]$$

$$= 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}$$

These positions are with minimum displacement, velocity, amplitude, acceleration but maximum strain are called **Nodes**.

Case-2

When, $\sin \frac{2\pi}{\lambda} x = \pm 1$ then, $\cos \frac{2\pi}{\lambda} x = 0$;

then, $y = \max$, $u = \max$, $a = \max$, $A = \max$

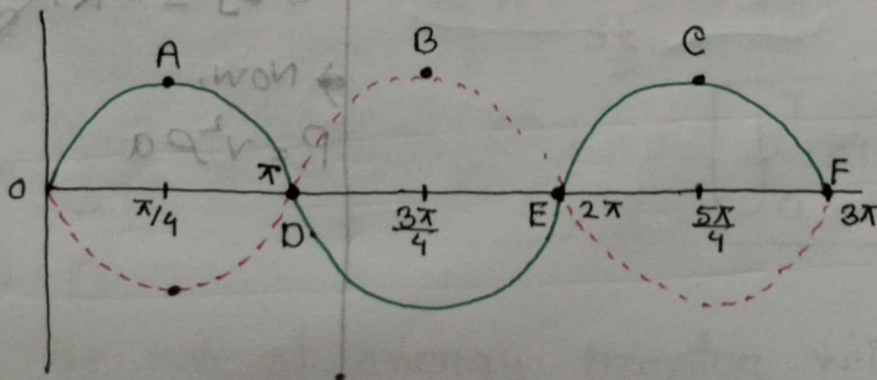
but $\frac{dy}{dx} = \text{minimum} \approx 0$

It is only possible when,

$$x = (2m+1) \frac{\lambda}{4} \quad [m = 1, 2, 3, \dots]$$

$$= \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots$$

These positions are with maximum displacement, velocity, acceleration, amplitude but minimum strain are called **anti-node**.



A, B, C
→ anti-node,

D, E, F
→ Node.

Energy transfer in case of stationary wave:

The equation of time energy after interval of time

dt,

$$= \int_0^T P u dt$$

$$= \int_0^T \frac{F}{A} \frac{dy}{dt} \cdot dt$$

$$= \int_0^T \frac{F}{A} dy$$

$$\left| \begin{array}{l} p = \text{pressure} \\ = F/A \end{array} \right.$$

and

u = particle velocity

→ ~~also~~ actually this indicates the work done per unit area.

rate of the energy transfer $= \frac{\int_0^T P_u dt}{T}$

Let, $U = U_x \sin \frac{2\pi vt}{\lambda}$

No, we know,

Bulk modulus and, $v = \sqrt{\frac{k}{\rho}}$

$$k = \frac{-P}{\frac{dy}{dx}} \Rightarrow k = v^2 \cdot \rho$$

$$\Rightarrow P = -k \frac{dy}{dx}$$

$$\Rightarrow P = v^2 \rho \cdot \frac{4\pi a}{\lambda} \cos \frac{2\pi}{\lambda} x \cdot \cos \frac{2\pi vt}{\lambda}$$

$$\Rightarrow P = P_x \cos \frac{2\pi vt}{\lambda}$$

So, the rate of the energy transfer :

$$= \frac{1}{T} \int_0^T P_x \cos \frac{2\pi vt}{\lambda} U_x \sin \frac{2\pi vt}{\lambda} dt$$

$$= \frac{P_x U_x}{2T} \int_0^T 2 \sin \frac{2\pi vt}{\lambda} \cos \frac{2\pi vt}{\lambda} dt$$

$$= \frac{P_x U_x}{2T} \int_0^T \sin \frac{4\pi vt}{\lambda} dt$$

$$= \frac{P_x U_x}{2T} \times 0 \quad \left[\int_0^T \sin \frac{4\pi vt}{\lambda} dt = 0 \right]$$

$$= 0$$

So, the rate of energy transfer will be zero = 0

Beat of Sound :-

When two sound waves of nearly same frequency and amplitude overlap / superimpose each other, then the resultant amplitude sound alternate between hard and soft sound, this phenomena is a beat.

Freq. diff $= \omega_1 - \omega_2 \leq 10 \text{ Hz}$ तब, $\frac{\omega_1 - \omega_2}{2} = \omega$

then, Beat तैरि तब $\frac{\omega_1 - \omega_2}{2} = \omega \Rightarrow$

Let, $y_1 = a \sin \omega_1 t$

$$y_2 = a \sin \omega_2 t$$

So, resultant,

$$y = y_1 + y_2$$

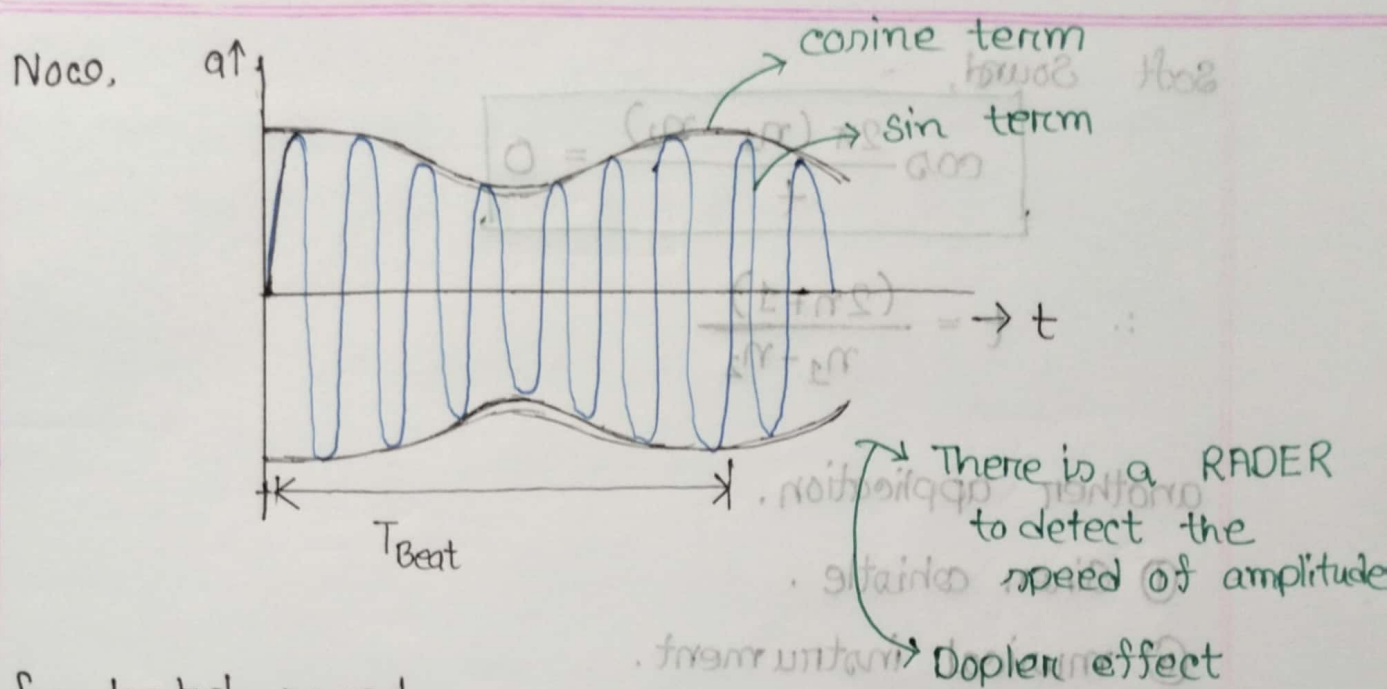
$$= a \sin \omega_1 t + a \sin \omega_2 t$$

$$= 2a \sin\left(\frac{\omega_1 + \omega_2}{2}t\right) \cos\left(\frac{\omega_1 - \omega_2}{2}t\right)$$

Modulated term $\leftarrow$

$$= \underbrace{2a \cos\left(\frac{\omega_1 - \omega_2}{2}t\right)}_B \sin\left(\frac{\omega_1 + \omega_2}{2}t\right)$$

$$= B \sin\left\{\left(\frac{\omega_1 + \omega_2}{2}t\right)\right\}$$



for haled sound,

$$N \cos \omega_1 t = \frac{n}{n_1 - n_2}$$

when amplitude maximum,

$$\cos \frac{2\pi(n_1 - n_2)}{2} t = \pm 1$$

$$\text{So, } t = 0, \frac{1}{n_1 - n_2}, \frac{2}{n_1 - n_2}, \frac{3}{n_1 - n_2}, \dots$$

Time interval between two beats,

$$T_{\text{Beat}} = \frac{1}{n_1 - n_2}$$

$$\text{Beat frequency} = (n_1 - n_2)$$

Soft Sound,

$$\cos \frac{2\pi (n_1 - n_2)}{t} = 0$$

$$\therefore t = \frac{(2n+1)}{n_1 - n_2}$$

another application,

(i) Siren whistle.

(ii) musical instrument.

(iii) to detect harmful gas in mine.

$$1 \pm \cos \frac{2\pi (n_1 - n_2)}{t} = 0$$

$$\frac{1}{n_1 - n_2} = T$$

$$(n_1 - n_2) = \text{beat frequency}$$

Photo-electric effect

PE-1

- * photo-electric effect - আলো surface থেকে metal.
- * wavelength of visible light 4000-7000 Å
- * Energy " " " 2-3 eV
- * photo-electric effect, Compton effect support particle theory of light.
- * Emission of e^- from metal surface illuminated by light on any other radiation.

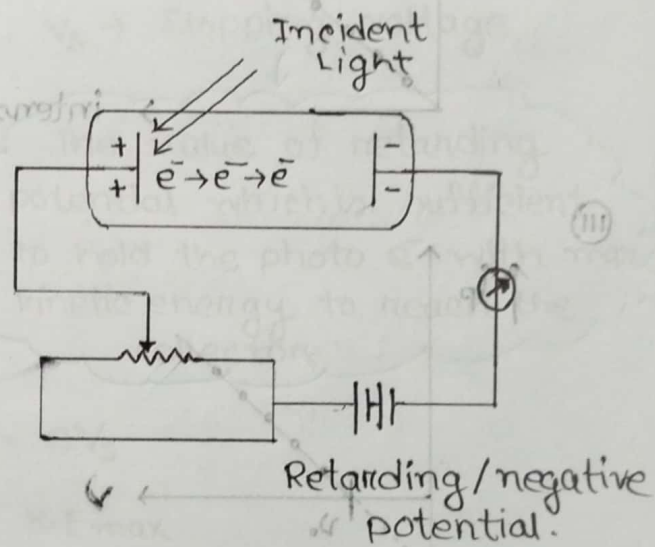
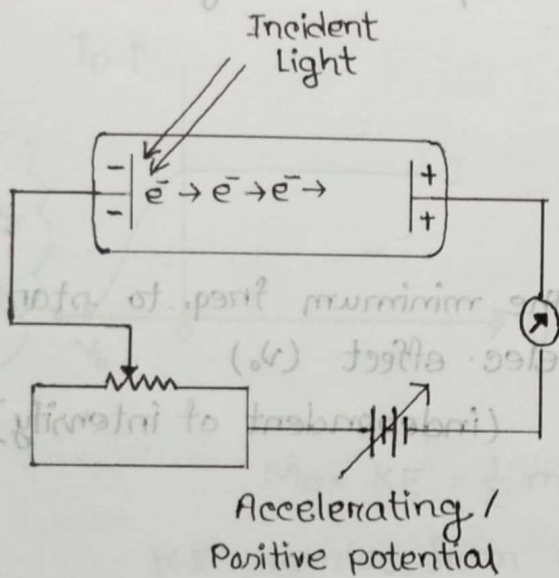
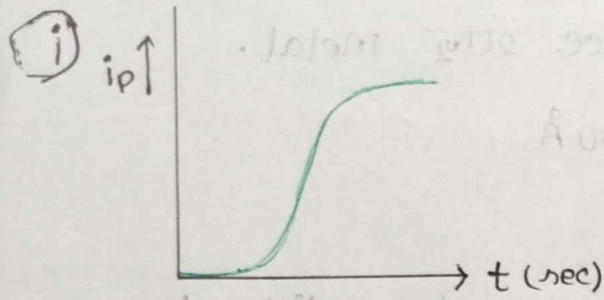


Photo-electric effect: when high frequency of light falls on the photo-sensitive metallic plane, then some e^- is emitted from it, this process is called photo electric effect. Electrons are called photo-electron and flow of current is called photo-current.

Threshold energy: Minimum energy is required to emit e^- .

Threshold frequency: " frequency " " " " e^- .

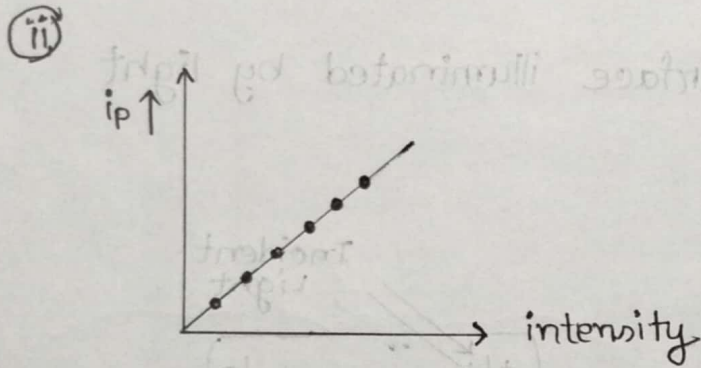
Experimental facts/laws:



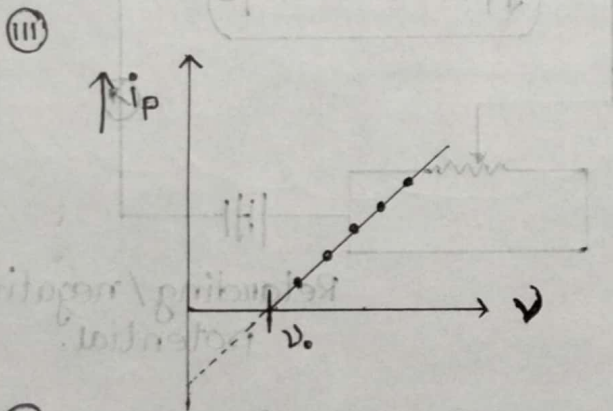
Time lags between the arrival of light on the metal surface and emission of electron.

← [Here no time lag between ---]

$i_p \rightarrow$ photo-current

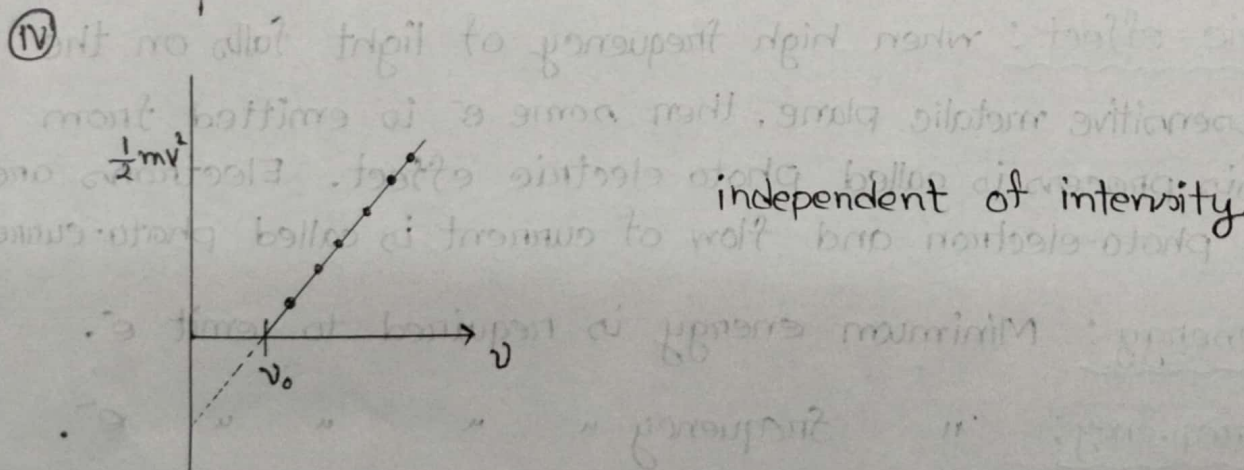


$i_p \propto \text{intensity}$

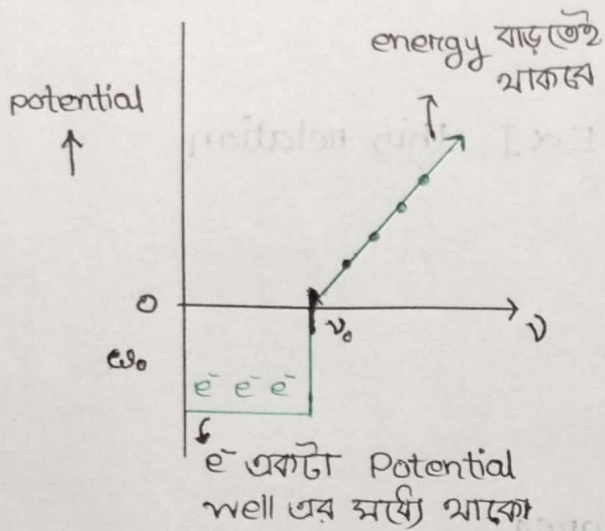


The minimum freq. to start photo-elec. effect (ν_0)

(independent of intensity)



(v)

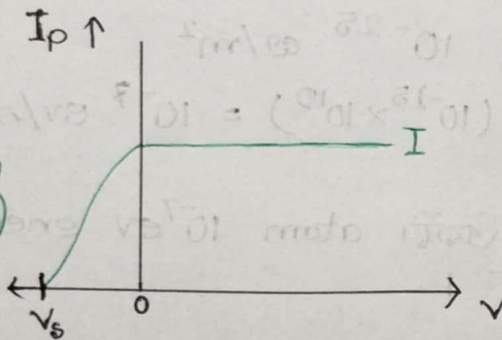


$$\frac{W_0}{\phi} = h\nu_0$$

Work function

Energy to get the e^- free from atom and away from the surface

(vi)



$v=0$, হলে কোনো battery না থাকলেও কিছু current পায়,

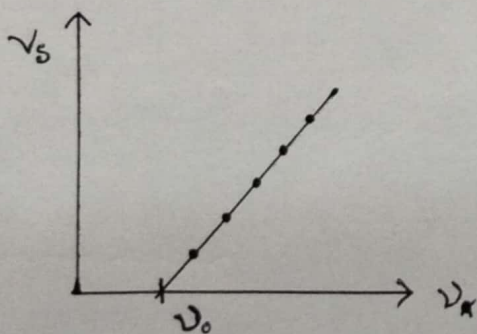
$v_s \rightarrow$ Stopping voltage

The value of retarding potential which is sufficient to hold the photo e^- with max kinetic energy to reach the collector

$$\text{Max KE} = \frac{1}{2} m v_{\text{max}}^2 = eV_s$$

KE varies from 0 to K.E_{max}

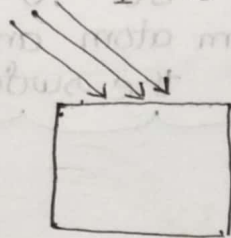
(vii)



Wave theory of light, $E \propto I \propto A^2$

But experimental data contradicts $E \propto I$ this relation

$$I = 10^{-10} \text{ W/m}^2$$



~ 1 atom thickness 1 m^2 area

10^{19} atom

Now, each atom will receive 10^{-25} W/m^2

So, $10^{19} \times 10^{-25} = 10^{-7} \text{ J/sec}$

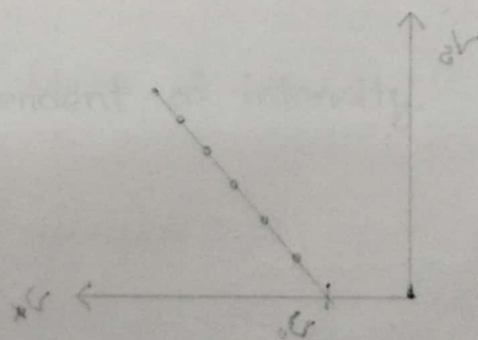
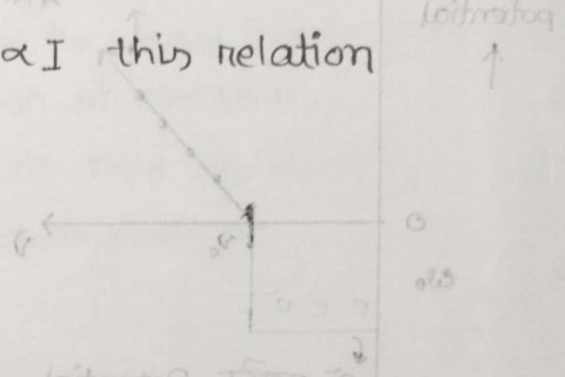
that means, প্রতি সেকেন্ডে একটি atom 10^{-7} eV energy accumulate করবে।

$W_0 = 1 \text{ eV} \Rightarrow$

if $t = 10^7 \text{ sec}$, theory says $t = 10^7 = 115 \text{ days}$

But experiment says $t = 0 \text{ s}$

Theory contradicts exp data.

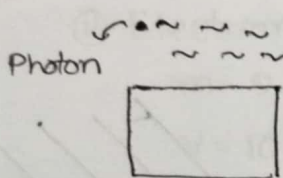


Max Planck $\rightarrow$ Quantum theory of Light :-

"Emission of light and radiation of energy is not a continuous process. It emits radiation of energy in discontinuous process like small packet called quantum. Light or any radiation is the summation of numerous radiation quantum and this quantum of light is called photon."

characteristics :

- 1) velocity is 3×10^8 in O medium
- 2) Uncharged and unscathed (নিষিদ্ধ)
- 3) Static mass is zero
- 4) momentum is $p = h/\lambda$

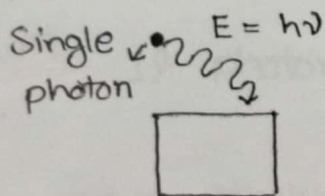


$$E = h\nu$$

Energy is quantized.

Each quanta is photon.

Einstein's Explanation of PEE of the basis of Q.T.L.



Particle to Particle interaction.

$$h\nu = \phi_0 + \frac{1}{2}mv^2$$

Einstein's P.E. effect equation

Hereby, Photo energy will be used as two purposes:

- (i) work function (ϕ_0 / ϕ)
- (ii) Kinetic energy ($\frac{1}{2}mv^2$)

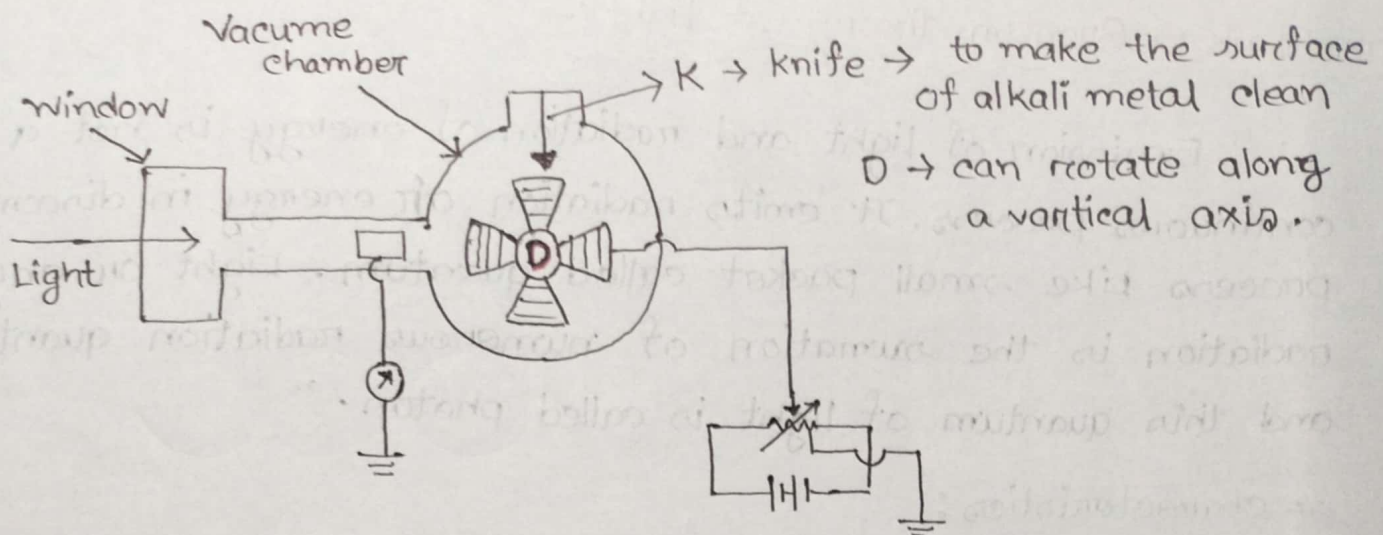
If total is work function, $h\nu = \phi_0 + 0$

$\rightarrow$ This explanation satisfies all the experimental facts.

**** verification of Einstein P.E.E by Prof. Robert milikan**

$\rightarrow$ target: to verify

$\rightarrow$ by determining v_s for diff materials at diff. freq.



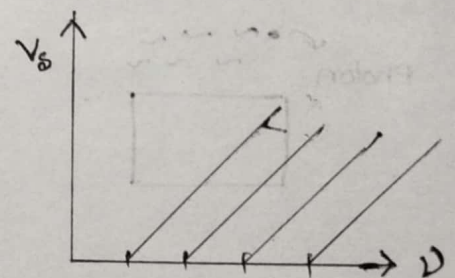
* At the begining of 20th century this experement was done.

→ in-situe → machine shop.
↓
in vacume

* He determined v_s for diff materials

$v_s \rightarrow K \rightarrow$ at diff freq.

$\rightarrow Na \rightarrow$ at ν



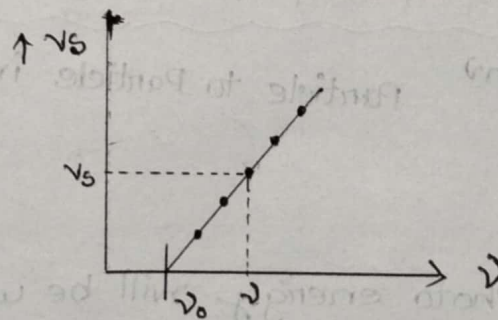
*
$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

$$\Rightarrow h\nu = h\nu_0 + eV_s$$

$$\Rightarrow eV_s = h(\nu - \nu_0)$$

$$\Rightarrow V_s = \frac{h}{e}(\nu - \nu_0)$$

$$\Rightarrow \frac{h}{e} = \frac{V_s}{\nu - \nu_0}$$
 slope of the, V_s vs ν graph



$$\Rightarrow h = e \times \text{slope}$$

$$\Rightarrow h = 6.65 \times 10^{-34} \text{ joule-sec}$$

which agrees with the standard value of 'h'

Hence, Experiment is ok.

Matter Wave

↳ neither sound wave
nor electromagnetic wave

* Louis victor de-Broglie proposed matter wave,
in his ~~Ph~~ Ph.D. Thesis (33 page) in 1924.

* The wavelength associated with a moving object of mass 'm' and velocity (v) is,

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

as example:

① Golf ball $\rightarrow m = 48g$
 $v = 50 m/s$

$$\therefore \lambda = \frac{6.63 \times 10^{-34}}{0.048 \times 50} = 288 \times 10^{-34} \text{ Hz}$$

↳ not measurable.

But

② Electron,
 $m = 9.11 \times 10^{-31} \text{ kg}$
 $v = 10^8 \text{ m/s}$

$$\therefore \lambda = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 10^8} = 7.3 \times 10^{-11} \text{ Hz}$$

↳ measurable.

* wave particle duality of Light
* " " duality of matter

Q. If photon (massless) acts as particle, why not moving e^- ?
mass-आहे.

Wave mechanics:

1. de Broglie
2. Higenberg
3. Schrodinger

$$P(x) = \int |\psi \psi^*| dx$$

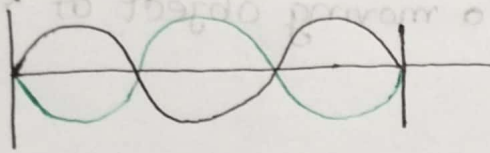
$$= \int |\psi \psi^*| ds \quad [\text{Surface}]$$

$$= \int |\psi \psi^*| dv \quad [\text{volume}]$$

Bohr Atomic Model:-

$$L = mvr = \frac{nh}{2\pi} \rightarrow \text{Bohr's quantization of atomic orbital.}$$

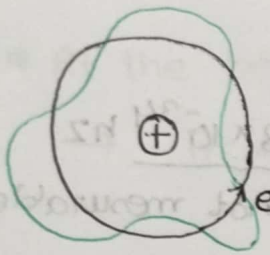
$$h\nu = E_2 - E_1$$



Standing wave

↓
Geometrical constraint

$$\frac{h}{mv} = \frac{h}{p} = \lambda$$



→ Geometrical constraint

$$2\pi r = n\lambda$$

$$\Rightarrow 2\pi r = n \cdot \frac{h}{mv}$$

$$\Rightarrow mvr = \frac{nh}{2\pi} \rightarrow \text{Bohr's quantization.}$$

that validates Bohr's model.

□

$$y = a \cos(\omega t - a)$$

$$v_p = \frac{\omega}{k} = n\lambda$$

$$\frac{d\omega}{dk} = v_g$$

phase velocity of de Broglie wave according to usual notation of wave motion,

$$v_p = n\lambda = \frac{nh}{mv} = \frac{E}{mv} = \frac{mc^2}{mv} = \frac{c^2}{v}$$

$$v_p > c \quad [\text{Phase velocity} > \text{velocity of light}]$$

→ It is not actual phase velocity,

→ It will be group velocity (v_g)

modulate velocity

$$\text{but } v_p = \overset{v}{\circ}$$

→ Particle velocity.

Group velocity :-

It is the velocity with which the overall shape of the wave amplitude, known as modulation or envelope propagate through space.

Let, $y_1 = a \cos(\omega t - kx)$
 $y_2 = a \cos[(\omega + d\omega)t - (k + dk)x]$

The resultant wave,

$$\begin{aligned} 2\omega + d\omega &= 2\omega \\ 2k + dk &= 2k \end{aligned}$$

$$y = y_1 + y_2$$

$$= a \cos(\omega t - kx) + a \cos[(\omega + d\omega)t - (k + dk)x]$$

$$= 2a \cos(\omega t - kx) \cdot \cos\left[\frac{d\omega}{2}t - \frac{dk}{2}x\right]$$

$$= \underbrace{2a \cos\left(\frac{d\omega}{2}t - \frac{dk}{2}x\right)}_{\text{weak part}} \cdot \underbrace{\cos(\omega t - kx)}_{\text{Strong Part}}$$

$$= 2a \cos\left[\frac{d\omega}{2}t - \frac{dk}{2}x\right] \rightarrow \text{This is modulated Part.}$$

$$= \cos \frac{1}{2} \left[dk \left(x - \frac{d\omega}{dk} t \right) \right]$$

$$v_g = \frac{d\omega}{dk}$$

$$\Rightarrow \cos \frac{1}{2} \left[dk (x - v_g t) \right]$$

$$\begin{aligned} v_g &= \frac{d\omega}{dk} = v_p + k \frac{\delta v_p}{\delta k} \\ &= v_p + k \frac{\delta v_p}{\delta \lambda} \cdot \frac{\delta \lambda}{\delta k} \end{aligned}$$

$$\begin{aligned} \therefore v_p &= \omega \lambda = \frac{\omega}{k} \\ \Rightarrow \omega &= v_p k \end{aligned}$$

$$v_g = v_p - \lambda \frac{\delta v_p}{\delta \lambda}$$

If phase velocity doesn't depend on λ , then

$$v_g = v_p \quad \text{cause} \quad \frac{\delta v_p}{\delta \lambda} = 0$$

Relation between group & Phase velocity

$$\begin{aligned} k &= \frac{2\pi}{\lambda} \\ \Rightarrow \lambda &= \frac{2\pi}{k} \\ \Rightarrow \frac{\delta \lambda}{\delta k} &= - \frac{2\pi}{k^2} \end{aligned}$$

আলো কোনা গিলাস-এর মতো দিয়ে গেলে we should use it

$$E^2 = m_0^2 c^4 + p^2 c^2$$

$$\Rightarrow 2E \cdot \frac{\delta E}{\delta p} = 2 p c^2$$

$$\Rightarrow \frac{\delta E}{\delta p} = \frac{2 p c^2}{2 E} = \frac{p c^2}{m c^2} = \textcircled{v} \rightarrow \text{object velocity.}$$

$$E = h\nu = \frac{h \omega}{2\pi}$$

$$\lambda = \frac{h}{p}$$

$$p = \frac{h k}{2\pi}$$

$$\frac{\delta E}{\delta p} = \frac{\delta \omega}{\delta k} = v_g$$

$$\therefore v_g = v$$

[Proved]

So, The de-Broglie's hypothesis is theoretically ok

Verification of matter wave

↳ Davison & Germer → Bell Lab

↳ G.P Thomson → U.K

↳ Son of J.J Thomson → discovered e^- .

EMF

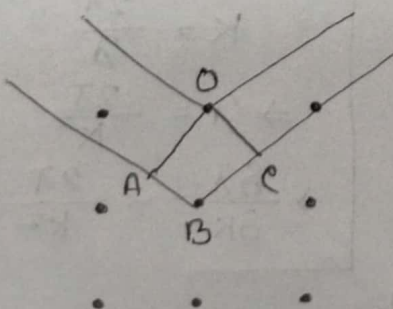
↳ Diffraction

↳ Interference

এই দুই Property Show করতে হবে ।

* X-Ray diffraction through crystal

* Bragg's law of x-ray diffraction

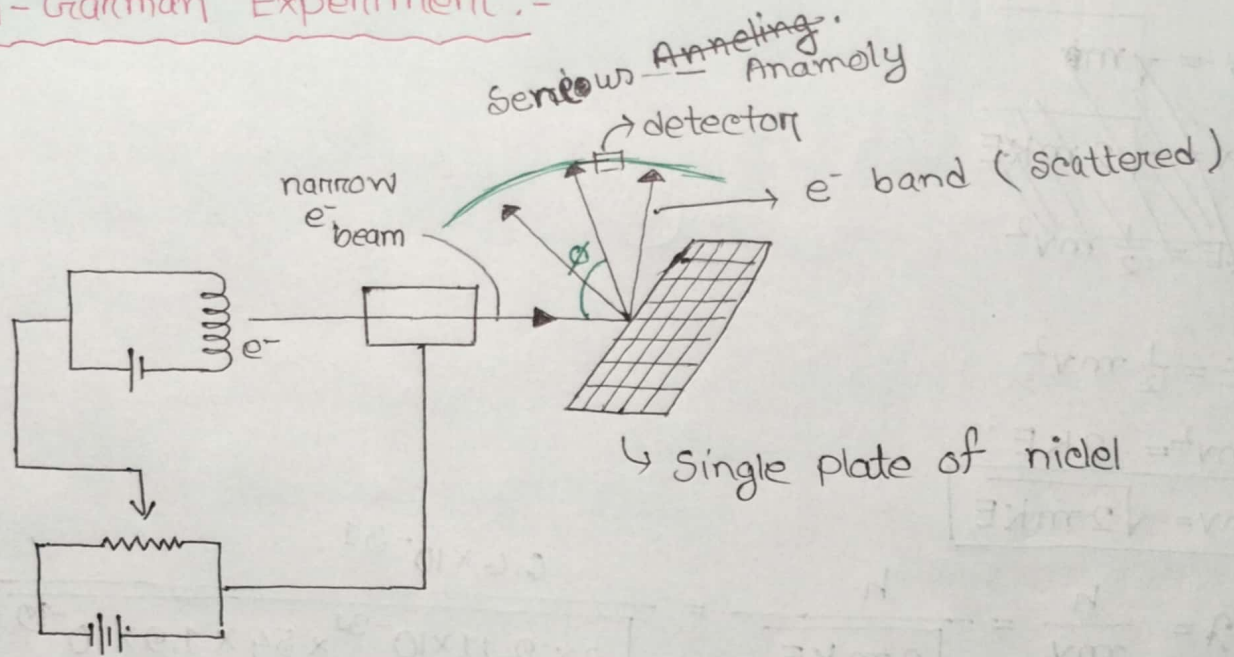


$$\text{Path diff} = AB + BC = 2d \sin \theta$$

$$\text{and } 2d \sin \theta = n\lambda$$

~~material at room temp~~

Davison - Garman Experiment :-

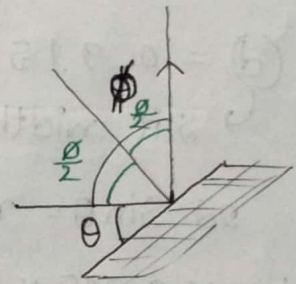
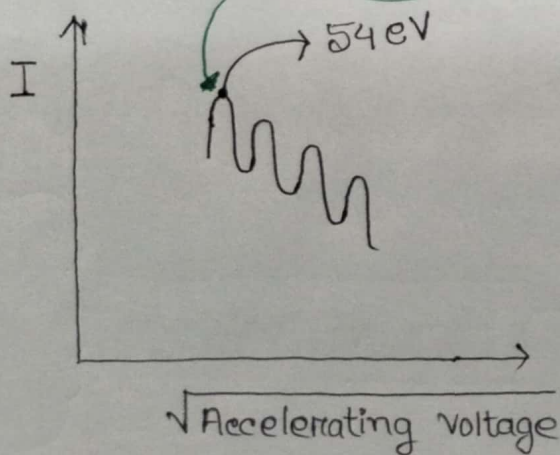


Annealing :- Room temperature থেকে বাড়িয়ে slowly room temperature -এ ফিড়িয়ে, আনা।

Now,

- * Davisson Anomally
- * Annealing:
- * e^- KE $\rightarrow$ 48-68 eV হলে,
- * যখন KE = 54 eV হলে,
তখন $\phi = 50^\circ$

Pronounced peak



$$\frac{\phi}{2} + \theta = 90^\circ$$

$$\text{Say, } \theta = 65^\circ$$

[wave particle duality is complete]

$$\rightarrow KE = \frac{1}{2} mv^2$$

$$\Rightarrow mv^2 = 2KE$$

$$\Rightarrow mv = \sqrt{2mKE}$$

$$\text{Now, } \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mKE}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 54 \times 1.6 \times 10^{-19}}} \\ \approx 1.66 \text{ \AA}$$

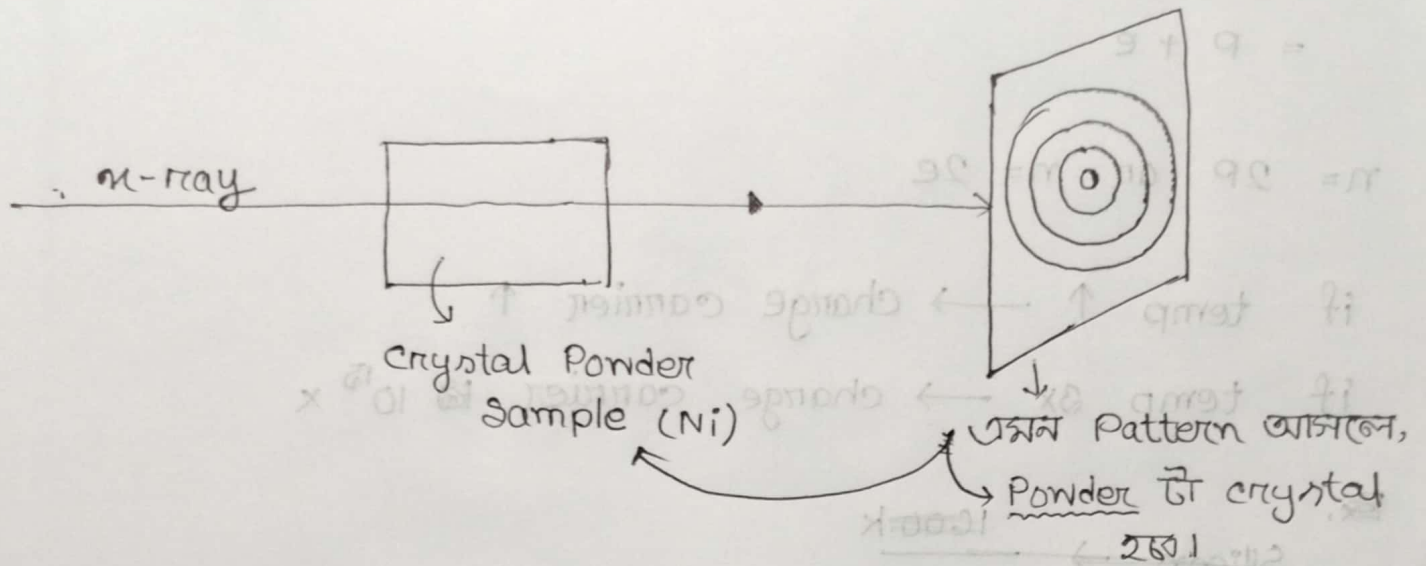
Ni $\rightarrow$ $d = 0.915 \text{ \AA}$
 $\hookrightarrow$ slit width.

$$2d \sin \theta = n\lambda$$

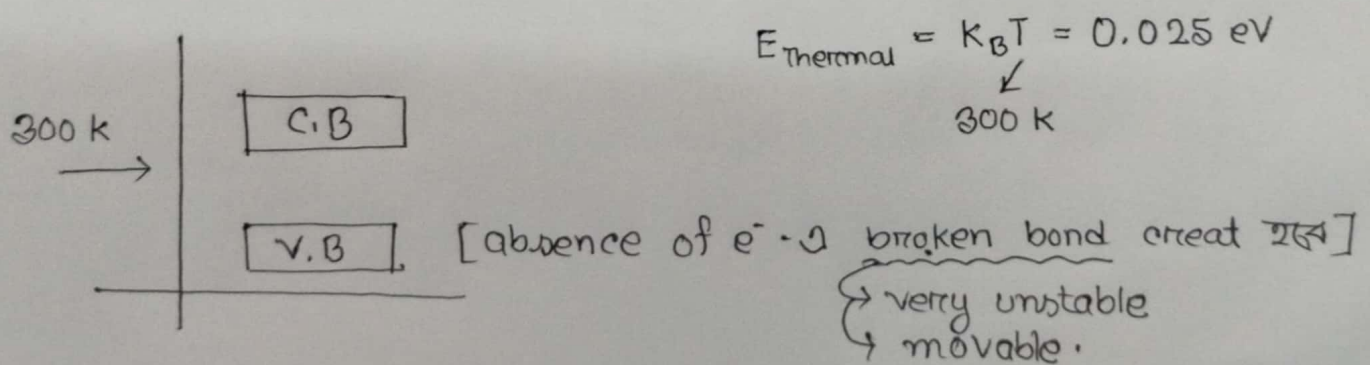
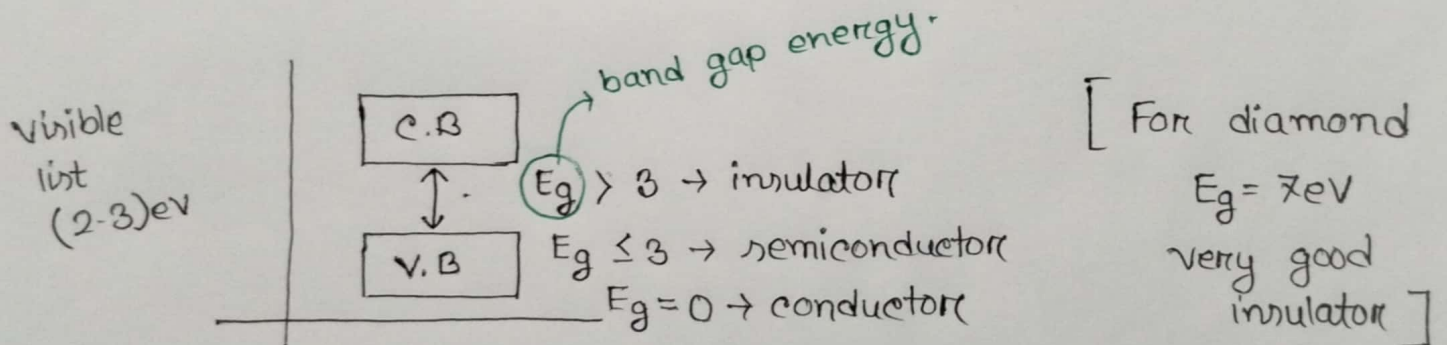
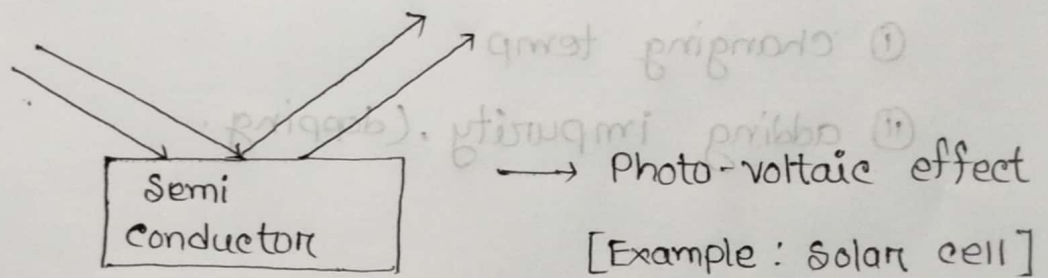
$$\Rightarrow 2 \times 0.915 \times \sin 65^\circ = n\lambda$$

[44-68 eV এর মধ্যে কোনো value.]

G.P. Thomson Experiment :



G.P. Thomson x-ray এর জায়গায় e-beam দিয়ে exp. করেন।



$$n = n_k + n_e$$

$$= p + e$$

$$n = 2p \text{ or } n = 2e$$

if temp $\uparrow \rightarrow$ charge carrier $\uparrow$

if temp $3\times \rightarrow$ charge carrier $10^{15} \times$

Ex, ~~1600K~~
Silicon $\rightarrow$

So, Semiconductor conductivity can be change in two ways

(i) changing temp

(ii) adding impurity, (dopping)